MILITARY SPECIFICATION

INSTALLATION, ENGINE

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification provides performance requirements and verification criteria for the integration of the propulsion system with the airframe. This specification applies to the following airframe portions of the engine installation subsystem developed for aircraft:

a. Air induction system
b. Cooling and ventilation
c. Engine exhaust system
d. Interfacing propulsion control mechanisms
e. Quick engine change assembly
f. Fire protection
g. Drains
h. Plumbing and wiring installation
i. Structural provisions
j. Accessibility/maintainability
k. Reliability
l. Vulnerability
m. Infrared signature and radar cross section
n. Noise
o. Environmental extremes
p. Interface control
q. Safety

1.2 Use. This specification cannot be used for contractual purposes without supplemental information.

1.2.1 Structure. The supplemental information required is identified by blanks within the specification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB, OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2840

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1.2.2 Instructional handbook. The instructional handbook which is contained in the appendix provides the rationale for specified requirements, guidance for inclusion of supplemental information, and a lessons learned repository.

1.3 Deviation. The details of any projected design for a given application which will result in improvement of system performance, reduced life cycle cost or reduced development cost through deviation from this specification or where the requirements of this specification result in compromise in operational capability, shall be brought to the attention of the procuring activity for consideration of change.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 **Propulsion installation.** The performance and functional capabilities of the integrated propulsion installation shall meet the requirements of this specification and those dictated by the operational environment and mission of the weapon system air vehicle as described in the air vehicle specification. The integrated propulsion installation shall provide a safe and compatible interface between the propulsion device and air vehicle for all operating conditions.

3.2 **Air induction system.** The air induction system shall consist of the following equipment and components: _____.

3.2.1 **Performance.** The air induction system shall provide a compatible interface, both internally and externally, between the propulsion system and the air vehicle under all ground and flight conditions specified in _____. The system shall provide total pressure variations, both instantaneous and discrete, at the engine compressor face within the limits established in _____ and defined in _____.

3.2.2 **Design and location.** The air induction system shall be designed and located to prevent engine ingestion of foreign objects, sand and dust, ice, armament gas and debris and other ground and airborne contaminants in excess of that permitted by _____.

3.2.2.1 **Foreign object damage (FOD) protection.** The air induction system shall be equipped with _____ to prevent engine ingestion of foreign objects.

3.2.2.2 **Sand and dust protection.** The air induction system shall be equipped with _____ to prevent engine ingestion of sand, dust and other erosive particles in excess of that defined by _____.

3.2.2.3 **De-icing systems.** A de-icing system shall be provided which, upon operation, will cause any accumulation of ice on the _____ to be removed.

3.2.2.4 **Anti-icing systems.** An anti-icing system shall be provided which will prevent the accumulation of ice on _____ when operating under the icing conditions specified in 3.2.3.

3.2.2.5 **Ice detectors and controls.** Aircraft inlet system noncontinuous deicing and anti-icing systems shall be provided with a manual and automatic control system. An ice detector shall be provided which will activate a visual warning device to alert the aircraft crew when icing of air inlet components occurs. The ice detector _____ have the capability to transmit a signal to initiate operation of the ice control systems.

3.2.3 **Icing environment.** The air induction system shall be capable of operating under the icing conditions specified by _____ throughout the engine ground and flight power range without the accumulation of ice on inlet system components that adversely affects engine operation or causes a loss of power or thrust.

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3.3 Cooling and ventilation. The engine compartment or nacelle shall be cooled and ventilated to prevent excessive engine accessory, equipment, supporting structure and compartment temperatures in excess of defined limits and the accumulation of explosive or flammable mixtures in or about the engine installation under any operating condition including the period after engine shutdown. Cooling system flow reversals or no-flow conditions shall not be permitted during any ground or flight condition; unidirectional airflow shall be provided under all conditions.

3.4 Engine exhaust system. The engine exhaust system shall include _____ _____.

3.4.1 Exhaust system function. The function of the exhaust system shall be to route the propulsion system exhaust gases and compartment cooling air directly and safely from the aircraft without damaging aircraft structure, creating a fire hazard, or FOD potential and _____.

3.4.2 Exhaust system cooling. Cooling or shielding shall be provided for the exterior surfaces of the engine exhaust system. Exhaust system surface temperatures shall not exceed _____°F, exposed surface temperatures in excess of _____°F shall be insulated.

3.4.3 Service life. The exhaust system service life shall not be less than _____ hours/cycles.

3.4.4 Special exhaust system equipment. The _____ shall be fail safe and upon failure shall not jeopardize safe operation of the aircraft or engine.

3.4.4.1 Thrust reversers. Thrust reversers shall incorporate the following safety features: _____.

3.5 Interfacing propulsion control mechanisms

3.5.1 Control mechanism function. The interfacing propulsion control mechanisms shall provide a safe and reliable method of transmitting the cockpit control lever(s) position to the engine contractor furnished controls. The power lever linkages between the flight station and the engines shall provide a linear kinematic relationship between the positions of the power levers in the flight station and at the engines.

3.5.1.1 Control linkage adjustment. Adjusting points shall be provided in the interface control linkages.

3.5.2 Cockpit control lever functional characteristics

3.5.2.1 Control levers in multiengine aircraft. The control levers in multi-engine aircraft shall occupy the same relative quadrant position for the same control setting throughout the entire engine power range.

3.5.2.2 Control quadrant detents and stops. The control quadrant shall have a detent at the _____ position(s). Control lever stops which require a deliberate motion to defeat shall be used at the _____ position(s).

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3.5.2.3 Friction lock. An adjustable friction device shall be incorporated in the throttle control mechanism.

3.5.3 Cockpit control lever performance

3.5.3.1 Control lever forces. The maximum force required to move the cockpit _____ shall not exceed _____ pounds per engine and _____ pounds for simultaneous movement of all levers.

3.5.3.2 Control lever deflection. The deflection of the _____ lever(s) while in the operating range shall not exceed _____ degrees in each direction when held rigid at any intermediate position and a force of _____ applied.

3.5.4 Turbojet and turbofan cockpit controls. A single cockpit power lever for control of each engine shall be installed in turbojet and turbofan powered aircraft which coordinates all engine variables through a single engine connection. The power lever quadrant shall be configured in accordance with figure 1, item A. Power lever angular travel shall _____.

3.5.4.1 Thrust reverser controls. A power lever quadrant in accordance with figure 1, item B, shall be provided for engine installations which incorporate thrust reversers. Thrust reverser controls shall be as follows: _____.

3.5.5 Turboprop cockpit controls. Two flight station controls for each engine-propeller combination shall be installed in turboprop aircraft; one engine power lever and one turboprop condition lever shall be provided. These controls shall have functional characteristics substantially in accordance with figure 1, items C and D.

3.5.5.1 Turboprop engine power lever. The power lever shall be in accordance with the standards shown on figure 2. In multiengine aircraft the power lever shall have a minimum vertical lift of _____. In single console installations (one set of power levers for both pilots) the maximum upward pull force shall not exceed _____ per lever. In dual console installations (one set of power levers for the pilot and a connected set for the copilot) the maximum upward pull force shall not exceed _____ per lever. The vertical lift line shall be at _____ to the plane of the pilot's seat. The upper _____ below the knob of each power lever shall be spaced as shown in the lower view on figure 2. The levers within this area shall have rounded edges and a thickness of _____.

3.5.5.2 Turboprop condition lever. The condition lever shall be in accordance with the standards shown on figure 3.

3.5.5.3 Reverse-pitch propeller controls. When reverse pitch propellers are installed, reverse pitch shall be obtained by movement of the engine power lever past the idle position in the aft direction through the lift point. Motion of the power lever shall energize the reversing circuits of the propeller control. Further travel in the aft direction shall increase reverse thrust by increasing engine power. The quadrant shall include a positive stop to prevent inadvertent reverse-pitch operation regardless of the landing gear position. The stop shall consist of an enclosed track having a vertical lift at the flight idle position and shall be bypassed by vertically raising the power lever. Movement of the lever into the aft or reverse position shall not

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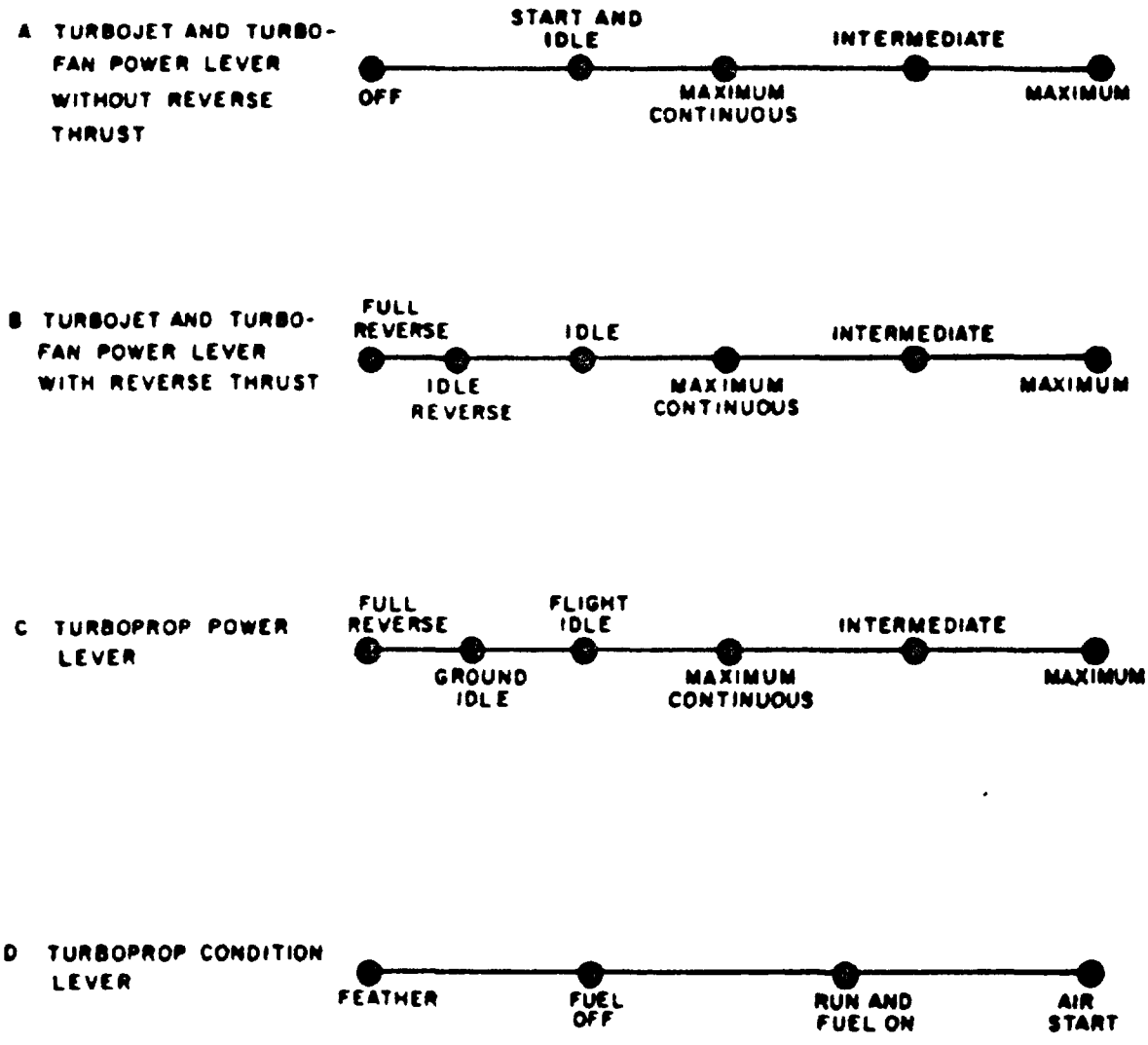
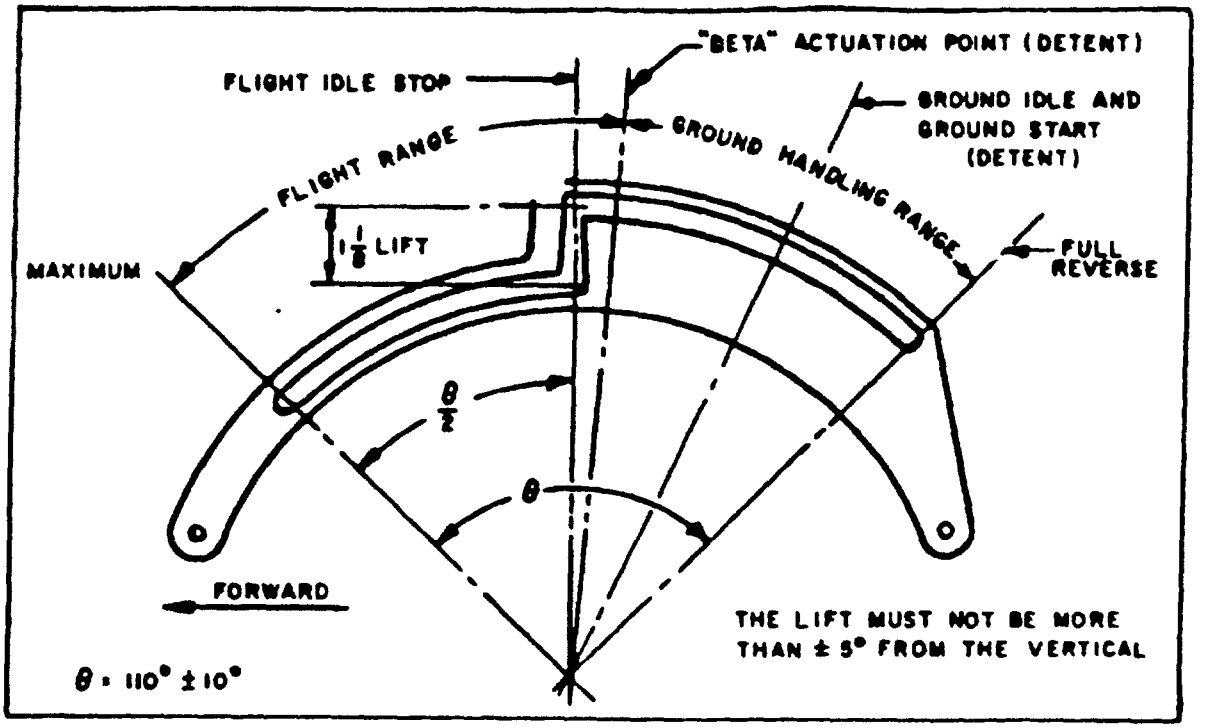
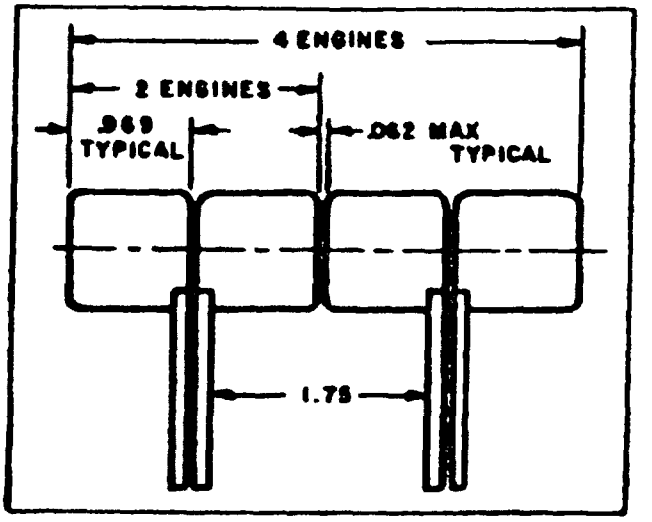


FIGURE 1. Turbine engine control quadrants.

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TURBOPROP POWER LEVER MECHANISM



TURBOPROP POWER LEVER ARRANGEMENT
(TWO OR MORE LEVERS)

LINEAR DIMENSIONS IN INCHES (BOTH VIEWS)

FIGURE 2. Turboprop power lever.

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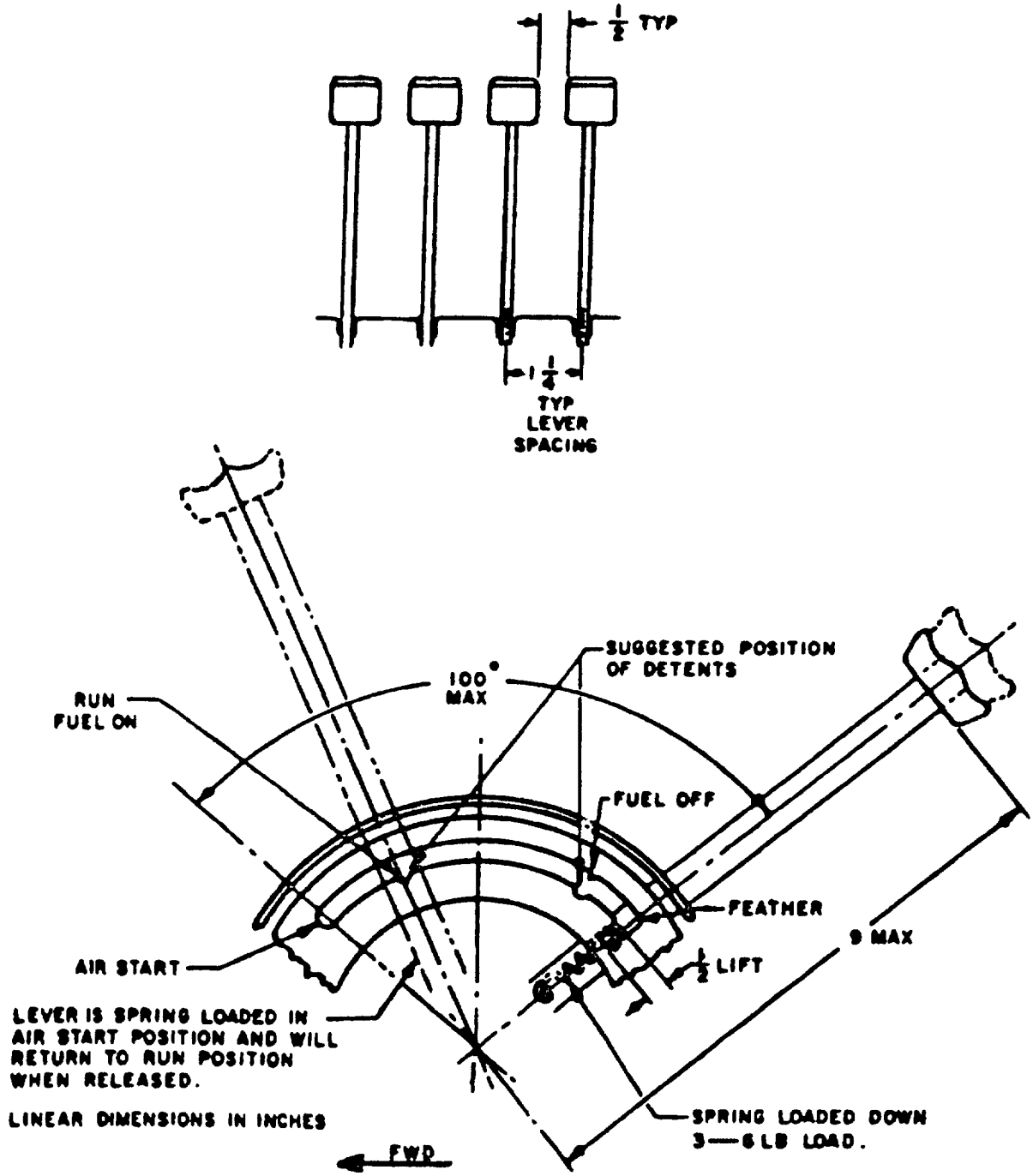


FIGURE 3. Turboprop condition lever.

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require removal of the operator's hand from the lever. Forward motion of the control shall be unrestricted and shall unreverse the propeller. In multi-engine aircraft equipped with two sets of controls, the power lever shall permit lift-to-reverse operation from either pilot's control. The controls shall permit reversal of any propeller independently of the others.

3.6 **Quick engine change assembly.** The quick engine change (QEC) assembly shall consist of the completely assembled propulsion system package with all components installed which can be readily transported for installation or removal from the aircraft. Components of the QEC, which include the engine and those equipments and accessories attached to or which form a part of the engine and its installation, shall be defined.

3.6.1 **Interchangeability.** Multiengine aircraft QECs shall be completely interchangeable between positions without physical or electrical modifications to any part of the QECs and without resorting to QEC sectioning.

3.6.2 **Removal and installation.** The time required to remove a QEC in operating condition and to install another QEC and put it in operating condition shall not exceed _____ minutes.

3.7 **Fire protection.** The engine installation shall be designed and protected from the hazards of fire in accordance with _____.

3.8 **Drains.** Closed compartments and equipment subject to fluid spillage or leakage shall be provided with drains for routing the fluid leakage or spillage overboard or to a collector system. The collector system shall have a capacity of _____.

3.9 **Plumbing and wiring.** Plumbing and wiring shall be installed in accordance with _____.

3.10 **Structural provisions.** The propulsion system installation shall be designed in accordance with the applicable structural design requirements of _____.

3.10.1 **Vibratory stress.** The installed propulsion system shall be free from destructive vibratory stress and wear.

3.11 **Accessibility/maintainability.** All equipment in the engine installation shall be located and mounted to permit access for rapid equipment testing, servicing, removal, inspection, and replacement using standard tools. The maintainability requirements of the engine installation designs and systems shall be compatible with overall aircraft maintainability requirements. The installation designs and systems shall give a maintenance manhour per flight hour (MMH/FH) no greater than _____.

3.12 **Reliability.** The reliability of the engine installation shall be compatible with the overall aircraft system. The mean time between failure (MTBF) for the engine installation designs and systems shall not be less than _____.

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3.13 **Vulnerability.** The vulnerability of critical propulsion components and accessories to threats commensurate with the aircraft mission shall be considered in the configuration of the propulsion system installation.

3.14 **Infrared (IR) signature and radar cross section (RCS).** The installed propulsion system shall meet the allocated IR and RCS requirements of the air vehicle specification.

3.15 **Noise.** Near and far field noise levels for the installed propulsion system shall not exceed the levels specified by _____. Maximum noise exposure limits for Air Force ground personnel in terms of sound pressure levels and exposure times shall not exceed _____.

3.16 **Environmental extremes.** Environmental extremes in which the propulsion installation components and systems must operate satisfactorily, include the following

Natural Environment	Requirement
Altitude	
Temperature	
Salt fog	
Humidity	
Rain	
Dust	
Fungus resistance	

Induced Environment	Requirement
Gunfire vibration	
Shock	
Explosive atmosphere	
Acceleration	
Electromagnetic	

3.17 **Interface control.** The airframe and engine physical, functional and performance interfaces shall be defined and controlled through an Interface Control Document.

3.18 **Safety.** Safety requirements for the engine installation shall include the following: _____.

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4. VERIFICATIONS

4.1 Propulsion installation. The installed integrated propulsion system shall be tested to verify that the performance and functional characteristics of the propulsion installation meet the requirements of this specification and those dictated by the operational environment and mission of the weapon system. Sufficient testing shall be accomplished to verify that a safe and compatible interface between the propulsion device and air vehicle has been provided for all operating conditions.

4.2 Air induction system. The equipment and components that comprise the air induction system shall be verified by physical inspection.

4.2.1 Performance. The performance requirements of the air induction system shall be verfified by wind tunnel, ground and flight testing.

4.2.2 Design and location. The design and location of the air induction system shall be tested to verify the capability to prevent engine ingestion of objects, particles and contaminants in excess of that allowed by _____.

4.2.2.1 Foreign object damage (FOD) protection. The inlet system with _____ installed shall be tested to verify its capability to protect the engine from foreign objects and to demonstrate inlet/engine compatibility.

4.2.2.2 Sand and dust protection. The inlet system with _____ installed shall be tested to verify capability to protect the engine from sand, dust and other erosive particles and to demonstrate inlet/engine compatibility.

4.2.2.3 De-icing systems. The de-icing system shall be tested to verify ice removal capability.

4.2.2.4 Anti-icing systems. The anti-icing system shall be tested to verify capability to prevent the formation of ice on protected surfaces.

4.2.2.5 Ice detectors and controls. Inlet system de-icing and anti-icing system controls shall be verified by inspection and demonstration. Detector capability to detect ice accretion, alert the aircraft crew, and activate ice protection systems shall be verified by a demonstration and test of the installed system.

4.2.3 Icing environment. The air induction system shall be tested to verify capability to operate in the specified icing environment without the accumulation of ice on inlet system components that could adversely affect engine operation or cause a loss of power or thrust.

4.3 Cooling and ventilation. The cooling and ventilation system's capability to prevent temperatures in excess of the defined limits and the accumulation of explosive or flammable mixtures in or about the engine installation shall be verified by _____.

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4.4 Engine exhaust system. Components of the engine exhaust system shall be verified by inspection.

4.4.1 Exhaust system function. The functions of the exhaust system shall be verified by ground and flight tests.

4.4.2 Exhaust system cooling. The exhaust system cooling design shall be verified by testing throughout the engine operating envelope.

4.4.3 Service life. The exhaust system service life shall be verified by test.

4.4.4 Special exhaust system equipment. Fail safe operation of the _____ _____ shall be demonstrated by test.

4.4.4.1 Thrust reversers. Thrust reverser safety features shall be verified by demonstration.

4.5 Interfacing propulsion control mechanisms

4.5.1 Control mechanism function. The functional capability of the interfacing propulsion control mechanisms shall be verified by test of the integrated propulsion control system.

4.5.1.1 Control linkage adjustment. Interface control linkage adjustment points shall be verified by inspection.

4.5.2 Cockpit control lever functional characteristics

4.5.2.1 Control levers in multiengine aircraft. The relative quadrant position of the control levers in multiengine aircraft shall be verified by inspection.

4.5.2.2 Control quadrant detents and stops. Control quadrant detents and stops shall be verified by inspection and demonstration.

4.5.2.3 Friction lock. The incorporation of an adjustable friction lock device shall be verified by inspection and demonstration.

4.5.3 Cockpit control lever performance

4.5.3.1 Control lever forces. The cockpit control lever force requirements shall be verified by test.

4.5.3.2 Control lever deflection. Control lever deflection shall be measured with the specified force applied to verify compliance with 3.5.3.2.

4.5.4 Turbojet and turbofan cockpit controls. A single cockpit power lever which coordinates all engine variables through a single engine connection shall be verified by inspection. Power lever angular travel shall be verified by test.

4.5.4.1 Thrust reverser controls. Thrust reverser controls shall be verified by inspection and demonstration.

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4.5.5 **Turboprop cockpit controls.** Flight station controls and their functional characteristics shall be verified by inspection and demonstration.

4.5.5.1 **Turboprop engine power lever.** The physical configuration and functional force limitations shall be verified by inspection and test.

4.5.5.2 **Turboprop condition lever.** The physical dimensions and functional characteristics of the condition lever shall be verified by inspection and test.

4.5.5.3 **Reverse-pitch propeller controls.** The design, operation, and function of the reverse-pitch propeller controls shall be verified by demonstration and test.

4.6 **Quick engine change assembly.** The quick engine change (QEC) assembly shall be inspected and the equipment and accessories which compose the QEC verified.

4.6.1 **Interchangeability.** Interchangeability of the QEC between the several locations on the aircraft shall be demonstrated, including determination of the normal time, personnel, and support requirements for removal and complete reinstallation.

4.6.2 **Removal and installation.** A QEC in operating condition shall be removed and replaced to an operating condition in _____ minutes.

4.7 **Fire protection.** The fire protection system shall be verified in accordance with the applicable verification requirements of _____.

4.8 **Drains.** Analyses shall be completed to determine that drain line sizes, pressure gradients, and flow conditions are compatible for proper drain system function relative to the propulsion equipment and installation. Propulsion installation system drains which, due to their configuration or location, type and quantity of fluid drained, etc., appear questionable, insofar as adequate drainage is concerned, shall be subjected to drainage tests. The drainage tests shall be conducted with dyed fluids and shall verify that the performance of all questionable and potentially hazardous drains is acceptable at all applicable aircraft ground and in flight operating attitudes and conditions. The capacity of the collection system shall be measured.

4.9 **Plumbing and wiring.** The plumbing and wiring installation shall be verified in accordance with the requirements of _____.

4.10 **Structural provisions.** Structural design requirements applicable to the propulsion system installation shall be verified by the methods specified in _____.

4.10.1 **Vibratory stress.** Stress surveys shall be conducted, as specified for the weapon system, to determine freedom from destructive stress or vibration in components of the installed propulsion system.

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4.11 Accessibility/maintainability. The ready accessibility for the purposes of inspection, cleaning, and adjustment (as well as removal and replacement) of the various propulsion system components using standard tools shall be demonstrated with the engine installed.

4.12 Reliability. The following analysis and test shall be accomplished to verify that allocated reliability requirements have been met.

TESTS	TEST HRS AND CYCLES	DEMONSTRATED RELIABILITY LEVELS

4.13 Vulnerability. The survivability of the propulsion system installation shall be verified by _____.

4.14 Infrared (IR) signature and radar cross section (RCS). The propulsion system IR signature and radar cross section shall be determined by test.

4.15 Noise. Near and far field noise tests shall be conducted on the installed propulsion system. Compliance with the specified noise exposure limits for ground personnel shall be verified by measurement.

4.16 Environmental extremes. All components within the propulsion installation system shall be subjected to the following environmental tests: _____.

4.17 Interface control. Definition and control of the airframe and engine physical, functional and performance interfaces shall be verified by inspection.

4.18 Safety. The safety requirements for the engine installation shall be verified by.

REQUIREMENT	VERIFICATION METHOD

5. PACKAGING

5.1 All deliverable items shall be prepared for shipment as directed by the procuring activity.

6.0 NOTES

6.1 Intended use. The engine installation covered by this development specification will integrate/interface the propulsion system with the airframe.

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6.2 **Responsible engineering office.** The office responsible for development and technical maintenance of this document is ASD/ENFPJ. Requests for additional information or assistance with this standard can be obtained from Mr. J. E. Wafzig, ASD/ENFPJ, Wright-Patterson AFB, OH 45433-6503, AUTOVON 785-4725, Commercial (513) 255-4725. Any information relating to Government contracts must be obtained through contracting officers.

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

Project No. 2840-F624

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APPENDIX

INSTALLATION, ENGINE

HANDBOOK FOR

10. SCOPE

10.1 Scope. This appendix provides rationale, background criteria and guidance, lessons learned, and instructions necessary to tailor sections 3 and 4 of the basic specification (MIL-I-87239) for a specific application.

10.2 Purpose. This appendix provides information to assist the Government procuring activity in the use of MIL-I-87239.

10.3 Use. This appendix is designed to assist the project engineer in tailoring MIL-I-87239. The blanks of the basic specification shall be filled in to meet operational needs of the equipment being developed.

10.4 Format

10.4.1 Requirements/verification identity. Section 30 of this appendix parallels section 3 and section 4 of the basic specification; paragraph titles and numbering are in the same sequence. Section 30 provides each requirement (section 3) and associated verification (section 4) as stated in the basic specification. Both the requirement and verification have sections for rationale, guidance, and lessons learned.

10.4.2 Requirement/verification package. Section 30 of this appendix has been arranged so that the requirement and associated verification is a complete package to permit addition to, or deletion from, the criteria as a single requirement. A requirement is not specified without an associated verification.

10.5 Responsible engineering office. The responsible engineering office (REO) for this appendix is ASD/ENFPJ, Wright-Patterson AFB, OH 45433-6503. The individual who has been assigned the responsibility for this handbook is J. Wafzig, ASD/ENFPJ, Wright-Patterson AFB OH 45433-6503, AUTOVON 785-4725, Commercial (513) 255-4725.

20. APPLICABLE DOCUMENTS

20.1 References. The documents referenced in this appendix are not intended to be applied contractually. Their primary purpose is to provide background information for the Government engineers responsible for developing the most appropriate performance values (filling in the blanks) for the requirements contained in the specification proper.

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20.2 Avoidance of tiering. Should it be determined that the references contained in this appendix are necessary in writing an RFP or building a contract, excessive tiering shall be avoided by calling out only those portions of the reference which have direct applicability. It is a goal of the Department of Defense that the practice of referencing documents in their entirety be eliminated in order to reduce the tiering effect.

20.3 Government documents

SPECIFICATIONS

Federal

TT-P-59	Paint, Ready-Mixed, International Orange (Not for Residential Use)

Military

MIL-C-7413	Couplings, Quick Disconnect, Automatic Shutoff, General Specification for
MIL-C-7958	Controls, Push-Pull, Flexible and Rigid
MIL-A-8064	Actuators and Actuating Systems, Aircraft, Electromechanical, General Requirements for
MIL-D-8181	Detector, Ice, Air Intake Duct, Aircraft Engines and Airframe Systems, General Specification for
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-A-8893	Airplane Strength and Rigidity, Sonic Fatigue
MIL-A-9482	Anti-Icing Equipment for Aircraft, Heated Surface Type, General Specification for
MIL-S-25057	Screen System, Air Inlet, Aircraft Turbine Engine, General Specification for
MIL-C-25427	Coupling Assembly, Hydraulic, Self-Sealing, Quick Disconnect
MIL-H-25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-H-38360	Hose Assemblies, Tetrafluoroethylene, High Temperature, High Pressure, 300 PSI, Hydraulic and Pneumatic
MIL-E-38453	Environmental Control, Environmental Protection, and Engine Bleed Air Systems, Aircraft, General Specification for

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MIL-F-87154	Fuel Systems
MIL-F-87168	Fire and Explosion Hazard Protection Systems, Aircraft, General Specification for
MIL-E-87219	Electrical Power Systems
MIL-A-87227	Hydraulic Power Systems
MIL-E-87231	Engines, Aircraft, Turbojet and Turbofan

STANDARDS

Military

MIL-STD-203	Aircrew Station Controls and Displays: Assignment, Location, and Actuation of, for Fixed Wing Aircraft
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-250	Aircrew Station Controls and Displays for Rotary Wing Aircraft
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1530	Aircraft Structural Integrity Program, Airplane Requirements
MIL-STD-1793	Flight Performance, Air Vehicle
MS 33602	Bolts, Self-Retaining, Aircraft Reliability and Maintainability, Design and Usage Requirements for
MS 33611	Tube Bend Radii
MS 33790	Bend Data, Minimum, Hydraulic, Pneumatic, Fuel and Oil Hose

20.4 Other publications

Federal Aviation Administration

FAR Part 25	Airworthiness Standards: Transport Category Airplanes
FAR Part 36	Noise Standards: Aircraft Type and Airworthiness Certification

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Air Force Systems Command Design Handbooks

AFSC DH 1-3	Human Factors Engineering
AFSC DH 1-6	System Safety
AFSC DH 2-2	Aircraft Stations and Passenger Accommodations

Air Force Regulations

AFR 80-36	Civil Airworthiness Standards for US Air Force Transport Aircraft
AFR 161-35	Hazardous Noise Exposure

Society of Automotive Engineers, Inc.

ARP 1055	Fire Resistance and Fire Test Requirements for Fluid System Components
AIR 1419	Inlet Total-Pressure-Distortion Considerations for Gas-Turbine Engines
ARP 1420	Gas Turbine Engine Inlet Flow Distortion Guidelines

30. REQUIREMENTS AND VERIFICATIONS

3.1 Propulsion installation. The performance and functional capabilities of the integrated propulsion installation shall meet the requirements of this specification and those dictated by the operational environment and mission of the weapon system air vehicle as described in the air vehicle specification. The integrated propulsion installation shall provide a safe and compatible interface between the propulsion device and air vehicle for all operating conditions.

REQUIREMENT RATIONALE (3.1)

The installed integrated propulsion system should be capable of functioning satisfactorily under all aircraft design operating conditions.

REQUIREMENT GUIDANCE

The required performance and functional capabilities of the total integrated propulsion system should be assessed relative to the mission requirements of the air vehicle. The necessary installation features should be provided as required to satisfy the performance and functional requirements of the installed propulsion system.

REQUIREMENT LESSONS LEARNED

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4.1 Propulsion installation. The installed integrated propulsion system shall be tested to verify that the performance and functional characteristics of the propulsion installation meet the requirements of this specification and those dictated by the operational environment and mission of the weapon system. Sufficient testing shall be accomplished to verify that a safe and compatible interface between the propulsion device and air vehicle has been provided for all operating conditions.

VERIFICATION RATIONALE (4.1)

The necessary level of confidence in the engine installations ability to satisfy the specified performance and functional requirements can only be obtained by test.

VERIFICATION GUIDANCE

The contractor should prepare a complete ground and flight test plan covering all tests and procedures to be accomplished by the contractor to demonstrate that the propulsion installation meets all requirements of the weapon system. This test plan should include as a minimum, all subsystem and specialty area testing identified throughout section 4 of this specification. As many of the required propulsion subsystem tests as possible should be accomplished simultaneously to assure that test time is most effectively used.

Analyses, model testing, simulator and wind tunnel tests, flight plans, and procedures, as prerequisite to flight operation, should be completed prior to the initiation of propulsion flight tests. Approval of the procuring activity should be obtained before flight tests may commence.

An inspection as specified by the procuring activity should be conducted prior to flight.

This will normally be a part of first flight certification for the aircraft. Flight release should be established according to the specific instructions of the procuring activity.

The contractor should obtain sufficient test data to adequately determine the performance characteristics of propulsion installation. Any part of the propulsion installation which fails its respective test or fails to function during any test should be rejected and require correction and retest. Propulsion system installation tests should be considered complete when the test data and analyses show that the installation meets all requirements; the operating limits of the equipment are not exceeded throughout the aircraft operating envelope; operating characteristics are stable, responsive, and show full compatibility among systems; performance data are satisfactory and do not reveal objectionable deterioration rates; and test hardware data and inspection do not reveal failure, pending failure, or undue wear.

VERIFICATION LESSONS LEARNED

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3.2 Air induction system. The air induction system shall consist of the following equipment and components _____.

REQUIREMENT RATIONALE (3.2)

This requirement defines the necessary airframe furnished ducts, scoops, passages or plenum chambers, controls, and equipment which provide the propulsion system with an air supply of the proper quantity and quality for combustion and cooling throughout the aircraft operational flight envelope.

REQUIREMENT GUIDANCE

The air induction subsystem equipment and components that have an influence on the engine air supply should be defined. Typical equipment and components that may be included in the air induction subsystem are:

a. Variable geometry actuator mechanisms, ramps, and doors.

b. Inlet protection systems such as inlet screens, inertia separators, vortex breakers, and inlet filters.

c. Ice detection, anti-icing and de-icing equipment.

d. Radar cross section reduction equipment.

e. Inlet noise suppression equipment.

f. Any equipment employed in or near the intake ducts which restricts, modulates, filters, heats or cools the intake air or supporting structure.

REQUIREMENT LESSONS LEARNED

4.2 Air induction system. The equipment and components that comprise the air induction system shall be verified by physical inspection.

VERIFICATION RATIONALE (4.2)

The equipment and components that comprise the air induction system are readily verified/identified by physical inspection.

VERIFICATION GUIDANCE

Subassemblies should be viewed before system integration and final assembly to facilitate physical inspection of equipment and components.

VERIFICATION LESSONS LEARNED

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3.2.1 Performance. The air induction system shall provide a compatible interface, both internally and externally, between the propulsion system and the air vehicle under all ground and flight conditions specified in _____ _____. The system shall provide total pressure variations, both instantaneous and discrete, at the engine compressor face within the limits established in _____ and defined in _____.

REQUIREMENT RATIONALE (3.2.1)

The performance of the air induction system significantly affects air vehicle and engine performance. Compliance with requirements of the specified documents is essential to insure airframe and engine compatibility.

REQUIREMENT GUIDANCE

Air vehicle ground and flight performance requirements are defined in MIL-STD-1793. Applicable requirements may also be defined in the air vehicle specification for the applicable program. Total pressure variations at the engine compressor face should be specified in the applicable engine specifications and defined in the interface control document. SAE documents AIR 1419 and ARP 1420 specify industry standard techniques for assessing inlet/engine compatibility.

REQUIREMENT LESSONS LEARNED

4.2.1 Performance. The performance requirements of the air induction system shall be verified by wind tunnel, ground and flight testing.

VERIFICATION RATIONALE (4.2.1)

The performance of the air induction system significantly affects air vehicle and engine performance. Induction system development experience has shown that the most effective means of verifying compatibility is a series of tests starting in the wind tunnel and continuing through flight testing.

VERIFICATION GUIDANCE

The applicable analysis, test, inspections and demonstrations of the specified documents applicable to the installed integrated propulsion system should be performed. The following conditions should apply to all tests of the air induction system:

a. The induction system should be in the flight configuration, operating to the required schedule.

b. Other features of the aircraft affecting air induction should be in the flight and mission configuration, operating to the required schedule. All features should be operated under the scheduled and critical range of conditions.

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c. All the various military loadings, including external weapons and stores configurations, that affect air induction performance should be used in specified arrangements and flight conditions including engine-out influence on inlet performance in dual engine/pod installations.

d. All auxiliary systems that extract air from the inlet or extract power or bleed air, or both from the engine, or in any way affect the engine tolerances to distortion should be operated normally during the test procedures that follow. Also, any auxiliary systems that in any manner affect the airflow at the inlet should be operated in a normal and failed mode.

e. Except where otherwise specifically noted, the specified tests should be conducted with the proposed variable geometry scheduling and airframe inlet duct configuration incorporated. Also, sufficient tests should be conducted to determine any effects of the variable geometry inlet being off-schedule.

As a minimum sufficient test data should be obtained through ground and flight testing to verify the performance and stability requirements of the complete induction system and to verify complete compatibility between the airframe and engine under all required operating conditions. The instrumentation and specific test data should accurately define the effects of the following variables where applicable:

a. Mach number and altitude -

b. Angle of attack

c. Angle of yaw (including crosswinds)

d. Ambient temperature

e. Variable inlet geometry, including control inputs due to engine acceleration or deceleration, control tolerances, and response limits

f. Compressor air bleed and horsepower extraction

g. Induction system boundary layer bleed and anti-icing operation effects

h. Missile/gun firing gas ingestion

i. External stores flow interference

j. Munition jettisoning/bomb-bay doors

k. Effects produced by engine throttle transients and afterburner operation, light-offs, or shutdowns

l. Effects of induction system operation on aircraft flying qualities, stability, and control

m. Acoustical loading consistent with the requirements of MIL-A-8893

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n. Recirculation of exhaust gases and debris

o. Inlet/nozzle orientation or position

p. Liquid water conditions

q. Anti-icing system setting; icing conditions

r. Aircraft-induced back-pressure effects on the engine.

VERIFICATION LESSONS LEARNED

3.2.2 Design and location. The air induction system shall be designed and located to prevent engine ingestion of foreign objects, sand and dust, ice, armament gas and debris and other ground and airborne contaminants in excess of that permitted by _____.

REQUIREMENT RATIONALE (3.2.2)

Engine ingestion of objects, particles and contaminants in excess of that which the engine has been designed to tolerate may result in rapid engine wear, damage or possible catastrophic failure. Engine performance and operability losses that result from wear and damage may degrade aircraft mission capabilities.

REQUIREMENT GUIDANCE

Engine ingestion of foreign objects, sand and dust, ice, armament gas and debris and other ground and airborne contaminants shall not exceed that allowed by the engine specification tailored from MIL-E-87231.

The air inlet should be located and positioned in an area where there is little probability of ingesting foreign objects including runway water, ice and debris thrown up by aircraft wheels or blown off the runway into the inlet during thrust reversal. High engine airflows may generate vortices at the entrance to the inlet system which can pick up runway water, ice and debris. This should be considered when locating inlets in close proximity to the ground. The relative locations of engine air inlets and armament stores should be considered so that the entrance of rocket and gunfire exhaust gases and debris into the inlet system may be prevented. The selected system location should be assessed to determine the types and quantities of objects, particles and contaminants that may potentially be ingested by the system in an operational environment. Applicable protective features should be provided when the anticipated ingestion exceeds that allowed by the engine specification. Aircraft operating restrictions that do not affect mission capabilities may also be employed. The use of operating restrictions should be closely coordinated with the weapon system user prior to adoption.

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REQUIREMENT LESSONS LEARNED

Air induction system location and design contributed to engine foreign object damage and operability problems on the following aircraft: The wing root inlet location on one particular aircraft was susceptible to inlet airflow generated vortices that picked up runway debris, water and ice which was ingested by the engine. The inlet system was also susceptible to gun gas ingestion due to the relative location of the armament and inlet.

Another aircraft with a wing root inlet location was susceptible to inlet airflow generated vorticies that carried ground debris into the engine.

On another aircraft the air induction system was located above and behind the aircraft wings and armament and was susceptible to ice shedded from the wings and gun gas ingestion.

The low fuselage centerline inlet on another aircraft generated inlet airflow vorticles that were capable of picking up runway debris, water and ice. A vortex destroyer system was subsequently installed to eliminate inlet flow generated vortices capable of carrying ground debris into the engine. The vortex destroyer system consisted of high pressure bleed air plumbed to the lower lip of the inlet and blown forward toward the ground.

The intake design on one aircraft with dual engine PODS had a valley or concave surface positioned above the inlet that literally directed ice and water into the inlet when the cowling anti-ice system was activated.

4.2.2 Design and location. The design and location of the air induction system shall be tested to verify the capability to prevent engine ingestion of objects, particles and contaminants in excess of that allowed by _____.

VERIFICATION RATIONALE (4.2.2)

Inlet system design and location protective capabilities would be extremely difficult to verify by methods other than test.

VERIFICATION GUIDANCE

The applicable engine specification taylored from MIL-E-87231 should be specified. Testing should be conducted in a natural environment when possible.

VERIFICATION LESSONS LEARNED

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3.2.2.1 Foreign object damage (FOD) protection. The air induction system shall be equipped with _____ to prevent engine ingestion of foreign objects.

REQUIREMENT RATIONALE (3.2.2.1)

Engine ingestion of foreign objects can adversely affect engine integrity and performance capability. A catastrophic engine failure could result from ingesting foreign objects.

REQUIREMENT GUIDANCE

Inlet protection for the engine should be provided based on the engine's specified sensitivity to foreign object damage. Potential engine damage resulting from the ingestion of foreign objects encountered in an operational environment should be assessed. Engine or airframe mounted retractable inlet screens or an equivalent FOD protection system should be provided when necessary to protect the engine. When retractable inlet screens are used, they should be aerodynamically shaped and designed in accordance with MIL-S-25057. The screen should be capable of preventing the entrance of a quarter-inch diameter sphere and strong enough to stop a .50 caliber cartridge case with an impact velocity of 300 ft./sec. for turbo-props and 400 ft./sec. for turbojets. Objects that are collected on the screen should be retained during extension and retraction. Screens should be automatically extended and retracted with the landing gear and a manual backup control should be provided. Inlet screens are highly susceptible to icing; therefore, when an inlet system ice detector is installed, a screen lockout should be provided in the retracted position when an icing signal is initiated or the anti-icing system is activated.

REQUIREMENT LESSONS LEARNED

4.2.2.1 Foreign object damage (FOD) protection. The inlet system with _____ installed shall be tested to verify its capability to protect the engine from foreign objects and to demonstrate inlet/engine compatibility.

VERIFICATION RATIONALE (4.2.2.1)

Verification of foreign object protective device capabilities, effects on engine performance, and contribution to inlet distortion can only be accomplished by test. Sufficient confidence in protective capabilities and effects on engine performance can not be established by other methods.

VERIFICATION GUIDANCE

The protective device specified in 3.2.2.1 should be tested for effectiveness and effects on installed engine performance. With the protective device installed inlet/engine compatibility testing over the range of airspeeds and

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altitudes where the device will be in position should be conducted. When retractable screens are used, testing should be performed to assess screen debris retention and icing tendencies, self-cleaning capability, strength and fatigue resistance.

VERIFICATION LESSONS LEARNED

3.2.2.2 Sand and dust protection. The air induction system shall be equipped with_____to prevent engine ingestion of sand, dust and other erosive particles in excess of that defined by _____.

REQUIREMENT RATIONALE (3.2.2.2)

Engine ingestion of excessive amounts of sand, dust and other erosive particles can cause accelerated engine wear and performance and stability loss.

REQUIREMENT GUIDANCE

Engine ingestion of sand and dust shall not exceed that specified by the engine specification tailored from MIL-E-87231. An assessment of aircraft mission requirements should be made to determine the potential levels of sand and dust that the engine may have to ingest. Protective equipment should be provided if levels exceed that allowed in the engine specification. Engines in aircraft which are designed to operate from unprepared or primitive grass or dirt landing areas are especially susceptible to erosive contaminants. Inertial separators, barrier filters or other inlet filtering concepts should be employed to minimize engine ingestion of sand, dust and other erosive particles.

The protective device selected should have an efficiency at a dust concentration of 0.0015 grams/cu. ft. of 80 percent or greater for particles of the 5-50 micron size and 100 percent for particles above 50 microns. The inlet system total pressure loss as a result of the protective device should not be greater than 0.5 percent and engine power or thrust loss should not exceed one percent during operation of the device. No power loss should occur when the device is in a passive or bypass mode. The protective device should be capable of satisfactory operation at all applicable engine airflows. The device should be self-cleaning and contain adequate ice protection if susceptible to ice accretion. A bypass air system to prevent excessive inlet pressure loss due to clogging or icing should be provided.

REQUIREMENT LESSONS LEARNED

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4.2.2.2 Sand and dust protection. The inlet system with _____ installed shall be tested to verify capability to protect the engine from sand, dust and other erosive particles and to demonstrate inlet/engine compatibility.

VERIFICATION RATIONALE (4.2.2.2)

Verification of sand and dust protective capabilities can only be accomplished by test. Sufficient confidence in protective capabilities and effects on engine performance cannot be established by other methods.

VERIFICATION GUIDANCE

The specified protective device should be tested for effectiveness and effects on installed engine performance.

VERIFICATION LESSONS LEARNED

3.2.2.3 De-icing systems. A de-icing system shall be provided which upon operation will cause any accumulation of ice on the _____ to be removed.

REQUIREMENT RATIONALE (3.2.2.3)

De-icing systems function to protect inlet system components from accumulating significant amounts of ice that could adversely affect engine performance and operation.

REQUIREMENT GUIDANCE

De-icing systems will allow the accumulation of ice on the protected surface and upon operation cause the accumulated ice to be removed from the surface. De-icing systems are typically cyclic and their rate of operation depends on the rate at which a hazardous quantity of ice accumulates on the protected surface. Airframe inlet components such as duct dividers, abrupt duct bends, boundary layer bleed wedges, bypass duct lips and engine duct accessory covers should as a minimum be de-iced.

REQUIREMENT LESSONS LEARNED

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4.2.2.3 De-icing systems. The de-icing system shall be tested to verify ice removal capability.

VERIFICATION RATIONALE (4.2.2.3)

Confidence in de-icing system capabilities can only be obtained by test.

VERIFICATION GUIDANCE

Hot air cyclic de-icing systems should be tested to verify compliance with the applicable portions of MIL-A-9482.

VERIFICATION LESSONS LEARNED

3.2.2.4 Anti-icing systems. An anti-icing system shall be provided which will prevent the accumulation of ice on _____ when operating under the icing conditions specified in 3.2.3.

REQUIREMENT RATIONALE (3.2.2.4)

Anti-icing systems offer the maximum protection to the engine from inlet icing induced performance losses and foreign object damage due to ice shedding.

REQUIREMENT GUIDANCE

The entrance to the air inlet system of all aircraft should be protected against ice formation. Anti-icing systems may operate continuously or intermittently. Systems which operate continuously should, if practical, use waste heat or heat that is normally lost. Systems whose operation results in a performance loss should operate only when an icing condition exists. However, an override should be considered if pilot selected manual operation is desired. An anti-icing system may run "wet" or run "dry" depending upon the surface being protected. A system which runs "wet" should maintain a surface condition which will melt ice, snow, and freezing rain which impinges on the surface and prohibit the resulting liquid from freezing on the surface. A system which runs "dry" should maintain a surface condition which will melt ice, snow, and freezing rain which impinges on the surface and will evaporate the remaining liquid. Aircraft inlet duct lips should incorporate a system designed to run "dry" as this system will normally prevent hazardous runback icing within the inlet duct.

REQUIREMENT LESSONS LEARNED

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4.2.2.4 Anti-icing systems. The anti-icing system shall be tested to verify capability to prevent the formation of ice on protected surfaces.

VERIFICATION RATIONALE (4.2.2.4)

Confidence in anti-icing system capabilities can only be obtained by testing.

VERIFICATION GUIDANCE

The inlet anti-icing system should be tested in natural or induced icing conditions to verify capabilities. Anti-icing system capability should be verified for the following ground and flight conditions:

With anti-icing system selector on:

a. Ground run time to simulate aircraft holding for takeoff clearance or taxiing long distances prior to takeoff.

b. Takeoff.

c. Maximum endurance power setting at 10,000 feet.

d. Best range power setting at 10,000 feet.

e. Maximum endurance power setting at 20,000 feet.

f. Best range power setting at 20,000 feet.

g. Simulated combat missions.

h. Idle decent from 20,000 feet.

i. Simulated ground control approach and touchdown.

VERIFICATION LESSONS LEARNED

3.2.2.5 Ice detectors and controls. Aircraft inlet system noncontinuous deicing and anti-icing systems shall be provided with a manual and automatic control system. An ice detector shall be provided which will activate a visual warning device to alert the aircraft crew when icing of air inlet components occurs. The ice detector _____ have the capability to transmit a signal to initiate operation of the ice control systems.

REQUIREMENT RATIONALE (3.2.2.5)

Manual and automatic control of ice control systems provides redundancy and flexibility. Aircrews may activate ice control systems in anticipation of icing conditions (i.e., prior to takeoff or landing). Ice detectors alert aircrews to icing conditions and facilitate passive crew control of inlet system icing protective devices.

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REQUIREMENT GUIDANCE

Ice detectors should conform to MIL-D-8181. The detector should be installed in the engine air inlet duct or in an area conductive to initial ice formation. The ice detector should be powered from an essential equipment bus and a checkout circuit should be provided which may be energized in flight so that a functional test of the detector probe and warning circuit may be accomplished.

REQUIREMENT LESSONS LEARNED

An ice detector was installed in an inlet location which was not sensitive to the high density water particles occurring during icing conditions. The flowpath of moisture heavy air should be considered when locating the sensor.

4.2.2.5 Ice detectors and controls. Inlet system de-icing and anti-icing system controls shall be verified by inspection and demonstration. Detector capability to detect ice accretion, alert the aircraft crew, and activate ice protection systems shall be verified by a demonstration and test of the installed system.

VERIFICATION RATIONALE (4.2.2.5)

Verification by the specified methods will provide a high degree of confidence in equipment capability to perform as specified.

VERIFICATION GUIDANCE

Ice detector response to ice accretion should be measured.

VERIFICATION LESSONS LEARNED

3.2.3 Icing environment. The air induction system shall be capable of operating under the icing conditions specified by _____ throughout the engine ground and flight power range without the accumulation of ice on inlet system components that adversely affects engine operation or causes a loss of power or thrust.

REQUIREMENT RATIONALE

Significant accumulations of ice on inlet system components can adversely affect engine operation and performance. Ice shed from inlet components constitutes a potential source of engine FOD or other airframe damage.

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REQUIREMENT GUIDANCE

Icing conditions are specified by the Federal Aviation Administration in FAR Part 25, Appendix C. MIL-E-38453 and MIL-E-87231 also specify icing conditions. Mission scenarios should be evaluated to determine if or what icing condition requirements are applicable.

REQUIREMENT LESSONS LEARNED

4.2.3 **Icing environment.** The air induction system shall be tested to verify capability to operate in the specified icing environment without the accumulation of ice on inlet system components that could adversely affect engine operation or cause a loss of power or thrust.

VERIFICATION RATIONALE (4.2.3)

Sufficient confidence in the air induction system's capability to operate in an icing environment can only be obtained by test.

VERIFICATION GUIDANCE

Testing should be performed in natural or simulated icing conditions. Ice detection, anti-icing and deicing systems incorporated in the inlet system should be tested to verify effectiveness.

VERIFICATION LESSONS LEARNED

3.3 **Cooling and ventilation.** The engine compartment or nacelle shall be cooled and ventilated to prevent excessive engine accessory, equipment, supporting structure, and compartment temperatures in excess of defined limits and the accumulation of explosive or flammable mixtures in or about the engine installation under any operating condition including the period after engine shutdown. Cooling system flow reversals or no-flow conditions shall not be permitted during any ground or flight condition; undirectional airflow shall be provided under all conditions.

REQUIREMENT RATIONALE (3.3)

Excessive engine compartment or nacelle temperatures can cause equipment damage and premature failure. The accumulation of explosive or flammable mixtures in or about the engine installation creates a hazardous situation for both the air vehicle and crew.

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REQUIREMENT GUIDANCE

The cooling and ventilation system should be of sufficient capacity to ensure that all established temperature limits within the engine installation are not exceeded under any operating condition. The airframe contractor should establish the maximum ambient temperature environments to which the engine and engine compartment plumbing, wiring, equipment, accessories, and structural components may be exposed during all periods of aircraft flight operation and ground operation, and after engine shutdown.

In establishing these limits, the airframe contractor should comply with the temperature limitations of the engine (as established by the engine manufacturer), engine components, and other installed equipment. In establishing post-shutdown temperature limitations, the contractor should consider that a gas turbine engine together with its usual installation orientation does not lend itself to pronounced convective circulation of cooling air. Post-shutdown cooling may further be restricted due to adverse operating conditions, such as blowing dust and salt spray, which require that all openings in the aircraft be covered as soon as possible after engine shutdown. Post-shutdown temperature limits should be satisfied without the need for auxiliary ground cooling equipment. Cooling and ventilating air intakes should be located so fuel, oil, and hydraulic fluid liquids and vapors and engine exhaust gases cannot enter the system. The air intake(s) location should not be susceptible to ice accretion. If ice formation is critical to an extent that airflow is adversely affected, the intake should have suitable ice protection.

The cooling system should not take air from the engine air inlet duct or plenum. In cases where this cannot be avoided, the cooling system should be designed to withstand the maximum pressure loadings resulting from an engine stall. Components of the cooling system in the engine air inlet duct should cause no significant inlet distortion and should not become a source of engine FOD. Air from other cooling and ventilation systems may discharge into the engine compartment(s) provided the air temperature is 200°F or less and cannot be contaminated with flammable, corrosive, or explosive agents which may result from normal or accidental leakage throughout any flight altitude or engine operating mode. Airflow used to cool and ventilate any engine compartment should be discharged overboard and should clear aircraft structure to minimize the chance of damage in the event of a compartment fire. Engine cooling and ventilating airflow should not be discharged into any other cooling and ventilation system except that used for exhaust nozzle cooling. Cooling air discharge from accessories such as oil coolers and generators should be discharged overboard.

When an engine compartment or nacelle is divided by one or more liquid and vapor barriers, separate cooling and ventilation systems should be provided for each compartment. There should be no air exchange between engine compartments.

REQUIREMENT LESSONS LEARNED

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4.3 Cooling and ventilation. The cooling and ventilation system's capability to prevent temperatures in excess of the defined limits and the accumulation of explosive or flammable mixtures in or about the engine installation shall be verified by _____.

VERIFICATION RATIONALE (4.3)

Verification of cooling system capabilities is necessary to assure that the engine installation environment is compatible with the temperature limitations specified for accessories, structure and other equipment of the installation. Proper ventilation of the installation must be assured to minimize hazards to the air vehicle, crew and ground personnel.

VERIFICATION GUIDANCE

The cooling and ventilation system should meet the requirements of 3.2.2 (as applicable to the cooling air inlet if it is susceptible to icing) as well as the interface requirements defined by 3.17. Determination of adequate engine/airframe/cooling system compatibility should be made by test, analysis and inspection of a flight configured installation.

Testing should be conducted to determine that the engine installation has satisfactory cooling characteristics under all required conditions of operation on the ground and in-flight. Applicable cooling data (skin temperature, airflow rates, airflow direction, and ambient temperature) should be obtained for the conditions tested for all individually cooled accessories and for each section of the engine (fan case, fan duct, compressor case, combustor case, turbine case, and exhaust duct). Sufficient data should be obtained to assure that unidirectional flow is present in the nacelle under all ground and flight conditions. The manufacturer's specified cooling requirements for each propulsion system component requiring cooling should be listed. Allowable operating temperature limits should be those specified in the applicable aircraft design specifications or those established (with Air Force approval) by the engine or component manufacturer.

Ground cooling tests should be conducted at ambient temperatures of at least 100°F and with the aircraft positioned crosswind with the wind velocity less than 10 knots. For multiengine installations, the test engine should be downwind of the fuselage. Maximum accessory drive load and maximum bleed air extraction should apply. Data should be recorded at the end of each of the run intervals and at 1-minute intervals or 30 minutes following engine shutdown. Data should include engine tilt attitude for V/STOL and in-ground effects when applicable. The engine should be stabilized for instrument readings at the following power levels observing any engine or propeller operating limitations.

a. Idle power.

b. 80 percent maximum continuous power. Include other power levels where compressor bleed is significant to nacelle or compartment temperature conditions.

c. Maximum continuous power.

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d. Intermediate power (or maximum nonaugmented).

e. Minimum augmented power (when applicable).

f. Idle approach.

g. Maximum augmented power (when applicable).

h. Idle power.

i. Perform transients as necessary to establish limiting temperature conditions and satisfactory characteristics.

Cooling and ventilation system flight tests should be performed at the following conditions. A takeoff and climb to the service ceiling altitude (or maximum allowable altitude within existing design limitations) should be conducted using maximum power and best rate-of-climb airspeed. Insofar as is practicable, data should be recorded at 5,000-foot intervals.

A takeoff and climb to the service ceiling altitude (or maximum allowable altitude within existing design limitations) should be conducted using intermediate power and best rate-of-climb airspeed. Insofar as practicable, data should be recorded at 5,000-foot intervals. If other takeoff and climb patterns are applicable to the aircraft, sufficient data should be taken to identify limiting conditions and required action taken. Level flight runs should be conducted at altitude intervals as required to show satisfactory cooling characteristics over the complete flight range. Each run should be for the time duration necessary to obtain structural and component temperature stabilization or for the maximum time within applicable limitations, whichever is shorter. Data should be recorded as necessary to show cooling and ventilation characteristics are satisfactory at several typical and worst case power conditions including the following:

a. Minimum aircraft speed.

b. Best duration flight speed.

c. Other applicable cruise speeds, including flight refueling speeds.

d. Maximum flight speed.

e. Maximum power including augmenter if applicable.

f. Minimum augmented power.

g. Where flow and cooling systems for multiengines are not fully independent, determine engine shutdown effects.

h. For turboprop, at maximum shaft horse power and lowest airspeed with loss of power on one engine if a two engine aircraft.

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Flights including all applicable maneuvers and combat runs should be conducted to determine that no adverse cooling, ventilation, or airflow condition occurs during any of these flight operations. All power ranges applicable should be used. Flight control effects should be determined, particularly those involving flow patterns in or around nacelles or engine bays.

The aircraft should be flown in various descent and landing patterns for determination that cooling and ventilation characteristics are satisfactory. For aircraft equipped with thrust reversers or reverse-pitch propellers, landing rolls should be performed using these devices at the worst conditions for determining that cooling, ventilating, and flow characteristics are satisfactory.

The following additional tests should be conducted for recording component temperatures, nacelle pressures, and airflow:

a. For STOL systems, maximum gross weight takeoff (in minimum distance) and climb to service ceiling should be conducted. Landing in minimum distance with maximum landing weight should also be conducted.

b. For VTOL systems, vertical takeoff at maximum permissible gross weight, hovering (in- and out-of-ground effect), and landing under varying crosswind conditions should be conducted. Transitions from vertical to level flight and from level to vertical flight should be conducted. Additional in-ground effect hover tests should be conducted for VTOL systems with derated (or flat rated) propulsion systems. In-ground effect hover tests should be conducted at various altitude elevations until the engine intermediate rated gas temperature is attained.

In addition to the ground and flight testing, sufficient data should be obtained to allow proper calibration of the fire/overheat detection system. Critical cooling such as may be associated with the use of secondary power units, heavy gross weight operation, in-flight refueling, engine or component malfunction, cooling airflow null points, etc., should be investigated as applicable. Operation of special features in the cooling system, such as suck-in doors, blowout doors or panels, controllable air inlets, and cooling air ejectors, should be demonstrated. The test procedures outlined herein are designed to provide a general survey from which further test conditions may be formulated based upon analysis of results and a knowledge of the aircraft systems involved. Prior to and during the required testing, it should be demonstrated that all of the general design requirements concerning cooling and ventilation have been met.

VERIFICATION LESSONS LEARNED

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3.4 Engine exhaust system. The engine exhaust system shall include _____ _____.

REQUIREMENT RATIONALE (3.4)

The components of the engine exhaust should be defined so that design responsibilities and interface requirements can be established.

REQUIREMENT GUIDANCE

The exhaust system should include those components which conduct the combustion gases from point of discharge at the engine to the point of discharge from the aircraft. The system should include the fan exhaust and the primary exhaust and all related components as installed by the airframe contractor, special exhaust system equipment such as thrust reversers, infrared suppressors, radar cross-section suppressors, noise suppressors, exhaust deflectors, etc., should also be included.

REQUIREMENT LESSONS LEARNED

4.4 Engine exhaust system. Components of the engine exhaust system shall be verified by inspection.

VERIFICATION RATIONALE (4.4)

Identification of engine exhaust system components is readily accomplished by inspection.

VERIFICATION GUIDANCE

An inspection of the airframe portions of the engine exhaust system should be made before and after installation of the engine.

VERIFICATION LESSONS LEARNED

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3.4.1 Exhaust system function. The function of the exhaust system shall be to route the propulsion system exhaust gases and compartment cooling air directly and safely from the aircraft without damaging aircraft structure, creating a fire hazard or FOD potential and _____.

REQUIREMENT RATIONALE (3.4.1)

The functions of the exhaust system should be identified to establish a baseline for design efforts.

REQUIREMENT GUIDANCE

The exhaust system may be used to reverse or vector engine thrust for enhanced aircraft control on the ground and in flight. The exhaust system should prevent exhaust gas impingement on the airframe and should also prevent ingestion of exhaust gases by any operating or nonoperating engine or any other air inlet where equipment or system operation, performance, or service life may be adversely affected. There should always be a positive pressure differential (higher pressure at the air inlet than at the exhaust outlet) during all flight conditions. The exhaust system should be compatible with the temperature and pressure environment associated with all engine operating conditions. The internal and external duct configuration should minimize restrictions to duct and cooling airflow. Operation of exhaust system equipment should not cause foreign objects to be ingested by the engines. Exhuast system components and equipment should be designed and constructed to resist corrosion. Where required, noise emitted and transmitted by the exhaust system should be attenuated.

REQUIREMENT LESSONS LEARNED

4.4.1 Exhaust system function. The functions of the exhaust system shall be verified by ground and flight tests.

VERIFICATION RATIONALE (4.4.1)

Ground and flight tests of the exhaust system are required to provide the necessary level of confidence in the functional capability of the exhaust system.

VERIFICATION GUIDANCE

As a minimum, the following data should be obtained during exhaust system ground and flight testing to determine that the exhaust system, including fan ducts, thrust reversers, and thrust vectoring systems, perform satisfactorily in the aircraft under all required conditions.

a. Ambient temperature and pressure.

b. Airflow parameters.

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c. Compressor air bleed and power extraction.

d. Power setting.

e. Fuel flow.

f. Position of variable geometry components as applicable.

g. Response rate of variable nozzle (if applicable).

h. Vibration characteristics of the system.

i. Temperatures of all surfaces subject to exhaust gas impingement.

j. Nozzle exit static pressure.

k. Jet plume velocities and temperatures during ground testing.

l. Additional data required by specific test procedures. Mach number, angle of attack, degrees of yaw, G load, and altitude should be obtained for all flight tests, in addition to above listed items as applicable.

The following data should be obtained during all tests of the thrust reverser system:

a. Time for thrust reverser deployment.

b. Time required for changing from deployed mode to stowed and locked mode.

c. Reverser effectiveness.

d. Ground plane efflux velocities.

e. Skin temperatures.

f. Loads on the significant reverser mechanisms.

g. Functional data required for analysis of all fail-safe features.

Sufficient ground tests should be performed to determine that performance of the engines with the nozzle and fan ducts is satisfactory, that there are no objectionable features or characteristics of operation, and that all required ground test data for evaluation are taken. During ground testing, the thrust reverser or vector nozzle should be deployed and stowed while the aircraft is moving at maximum and minimum taxi speed. If exhaust gas reingestion or adverse structural phenomena (such as high temperatures or vibration) are observed at minimum taxi speed, the reverser or vector nozzle should be deployed at various taxi speeds and nozzle angles until the minimum allowable speed for reverser or nozzle deployment is determined. Crosswind and tailwind conditions should be included in the test program. On VTOL/STOL systems with tilting inlets or vectoring nozzles, etc., all applicable conditions should be included in the ground test program. Engines should be run at all power

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levels and necessary inspections should be made to verify that the exhaust system meets all requirements and is ready for flight and to ensure that exhaust gases are not ingested by any inlet.

Flight tests should be conducted to verify operation of the exhuast system throughout the permissible flight envelope of the aircraft. Acceptable operation should be demonstrated for power levels from idle to maximum at all critical points within the permissible flight envelope. Takeoffs and climbs at all required power schedules and climb patterns to the operational ceiling of the aircraft should be conducted.

Level flights at varying altitudes, speeds, and power levels should be made to determine that exhaust system performance is satisfactory in level flight. Maneuvers and combat patterns applicable to the aircraft should be flown. For in-flight thrust reversing systems the thrust reverser should be deployed, stowed, and locked at the maximum and minimum allowable airspeeds and at various altitudes as necessary to determine the in-flight thrust reversing envelope and effectivity in aircraft maneuvers.

Thrust vectoring systems should be tested throughout the applicable range of speed and altitude. System failure should be simulated while the system is deployed to assure that fail-safe operation is provided.

VERIFICATION LESSONS LEARNED

3.4.2 Exhaust system cooling. Cooling or shielding shall be provided for the exterior surfaces of the engine exhaust system. Exhaust system surface temperatures shall not exceed_____°F; exposed surface temperatures in excess of_____°F shall be insulated.

REQUIREMENT RATIONALE (3.4.2)

The exhaust system is a potential flammable fluid ignition source. Cooling or shielding reduces the hazard potential of the exhaust system.

REQUIREMENT GUIDANCE

An ejector shroud in place of an insulation blanket is recommended for rejecting heat dissipated through the walls of the exhaust system. The ejector should provide uniform distribution of the cooling medium in order to obtain maximum cooling efficiency with minimum flow requirements. The outer surface of the exhaust shroud should be maintained at a temperature below 600°F under all operating conditions. The shroud should contain a drain system which shall remove unwanted fluids under all normal ground and flight attitudes; there should be no internal traps in which combustible fluids may collect.

REQUIREMENT LESSONS LEARNED

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4.4.2 **Exhaust system cooling.** The exhaust system cooling design shall be verified by testing throughout the engine operating envelope.

VERIFICATION RATIONALE (4.4.2)

The necessary level of confidence in the exhaust system cooling design can only be obtained by test during in-flight operation.

VERIFICATION GUIDANCE

Cooling tests should be run using temperature indicating paint and thermocouples to survey the exhaust system surface temperatures. Analytical methods should be used to consider extreme conditions that cannot be tested.

VERIFICATION LESSONS LEARNED

3.4.3 **Service life.** The exhaust system service life shall not be less than _____ hours/cycles.

REQUIREMENT RATIONALE (3.4.3)

The exhaust system service life is used to establish remove and replace instructions for overhaul and to assess compatibility with engine life.

REQUIREMENT GUIDANCE

Exhaust system components should not buckle, wrinkle, warp, separate, crack or burn-through during their service life. The service life should be based on the exhaust system applications and requirements. Service life should be design life of engine with repair.

REQUIREMENT LESSONS LEARNED

4.4.3 **Service life.** The exhaust system service life shall be verified by test.

VERIFICATION RATIONALE (4.4.3)

Due to the critical influence the exhaust system has on aircraft performance and safety a verification of life should be accomplished by testing to provide a high level of confidence in system integrity.

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VERIFICATION GUIDANCE

Accelerated mission testing should be performed on the installed exhaust system to evaluate design service life.

VERIFICATION LESSONS LEARNED

3.4.4 Special exhaust system equipment. The _____ shall be fail safe and upon failure shall not jeopardize safe operation of the aircraft or engine.

REQUIREMENT RATIONALE (3.4.4)

Improper operation of exhaust system equipment could adversely affect aircraft control, both on the ground and in flight. Fail-safe modes of failure are required to prevent damage or injury to aircraft or crews.

REQUIREMENT GUIDANCE

Failure modes of special exhaust system equipment such as thrust reversers, vectoring nozzles, infrared suppressors, radar cross-section suppressors, noise suppressors, exhaust detectors, etc., should be fail safe. Thrust reversers and exhaust deflectors or attenuators should be designed such that, in the event of a single failure, the device will remain in or assume the forward thrust position with no degradation in aircraft or engine performance.

REQUIREMENT LESSONS LEARNED

4.4.4 Special exhaust system equipment. Fail safe operation of the _____ _____ shall be demonstrated by test.

VERIFICATION RATIONALE (4.4.4)

Fail safe design features should be tested to thoroughly demonstrate safe operation and that no adverse affects result from failure.

VERIFICATION GUIDANCE

The equipment should be tested throughout the applicable range of speed and altitude.

VERIFICATION LESSONS LEARNED

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3.4.4.1 Thrust reversers. Thrust reversers shall incorporate the following safety features: _____.

REQUIREMENT RATIONALE (3.4.4.1)

Improper operation of thrust reversers on the ground or in flight could adversely affect aircraft control. Safety features are necessary to preclude damage to the aircraft and injury to the crew and ground personnel.

REQUIREMENT GUIDANCE

Positive means should be provided to prevent reverser actuation with the engine thrust control throttles in the forward thrust position. An automatic thrust reduction feature can be incorporated to reduce engine thrust if the reversers are inadvertently moved from the stowed position. A lock-out feature should be provided to prevent application of full forward or full reverse thrust until the reversing components have nearly reached their full travel position for forward or reverse thrust. An emergency system should be provided to move the reversers to the forward thrust or stowed position in the event of a malfunction in the normal control system. A positive means should be provided to retain the reversers in the stowed position. The thrust reversers should be fail safe in the forward thrust mode so that, in the event of an actuating mechanism failure during reverser thrust or modulated thrust operation, the doors will move them to a position yielding forward thrust.

REQUIREMENT LESSONS LEARNED

4.4.4.1 Thrust reversers. Thrust reverser safety features shall be verified by demonstration.

VERIFICATION RATIONALE (4.4.4.1)

Safety features of the thrust reversers should be demonstrated to establish the necessary level of confidence in their functional capability.

VERIFICATION GUIDANCE

Reverser safety features should be demonstrated.

VERIFICATION LESSONS LEARNED

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3.5 Interfacing propulsion control mechanisms

3.5.1 Control mechanism function. The interfacing propulsion control mechanisms shall provide a safe and reliable method of transmitting the cockpit control lever(s) position to the engine contractor furnished controls. The power lever linkages between the flight station and the engines shall provide a linear kinematic relationship between the positions of the power levers in the flight station and at the engines.

REQUIREMENT RATIONALE (3.5.1)

The functions of the interface mechanisms should be identified to establish a design baseline. A linear kinematic relationship between the cockpit thrust controls and the engine power lever is required to facilitate accurate and repetitive blind positioning of the throttle by the pilot.

REQUIREMENT GUIDANCE

The interfacing propulsion control mechanisms consist of those airframe furnished mechanical, pneumatic, hydraulic, and electric linkage mechanisms and related components which connect the cockpit control levers to the engine contractor furnished controls. The interfacing control mechanisms include, for example, those associated with turbojet, turbofan, and turboprop power levers (throttles) and propeller condition levers. The engine contractor furnished controls are not covered by this specification. The propulsion system cockpit controls should comply with MIL-STD-203 and MIL-STD-250.

The interfacing control mechanisms may consist of push-pull rods and bellcranks, cables and pulleys, flexible controls, remote-positioning controls, force amplifiers, hydraulic controls, or combinations thereof. The controls should be positively operated; springs should not be relied upon to actuate the linkage mechanism in either direction. Antibacklash devices may be used in the control systems; however, upon failure of the control system such devices should not cause the linkage mechanism to assume a position which will make the aircraft unsafe or uncontrollable. The control linkages should be provided with self-retaining bolt conforming to MS33602. Engine control linkages used in fire zones should be constructed of stainless steel or equivalent fire-resistant material. Design coordination is necessary between the airframe flexible cable designer and the engine fuel control designer in the mounting of the flexible cable pulley box on the fuel control input shaft to avoid introduction of excessive side loading and friction at the output terminal of the cable. Flexible and rigid push-pull controls, when used, should conform to MIL-C-7958 and should have backlash adjustments at one end of the control cable. Push-pull controls should have quick disconnects. Electromechanical actuators, if used, should be in accordance with MIL-A-8064. If electromotor actuators are used to actuate the engine fuel control input lever, limit switches should be provided to switch off current to the motor when driven against lever stops on the fuel control.

Force amplifiers, when used, should be fail safe in that an amplifier failure should result only in the loss of the force amplification feature; the integrity of the control should be maintained, and the resulting control actuating force should not exceed the maximum allowable by this specification.

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To eliminate dead band input to electronic engine controls, the power lever position potentiometer should be located on the cockpit throttle quadrant and not on the engine fuel control.

REQUIREMENT LESSONS LEARNED

4.5 Interfacing propulsion control mechanisms

4.5.1 **Control mechanism function.** The functional capability of the interfacing propulsion control mechanisms shall be verified by test of the integrated propulsion control system.

VERIFICATION RATIONALE (4.5.1)

The necessary level of confidence in the integrity and precision of the interfacing propulsion control mechanisms can only be obtained by testing.

VERIFICATION GUIDANCE

Testing of the integrated propulsion control system should be conducted under ground and flight conditions affording evaluation of the propulsion control interfacing mechanisms over the widest range of ambient temperature and weather conditions. Power lever control angles, engine control lever angles, and other control positions as pertinent to the test should be measured. Testing should be accomplished to assure integrity, accuracy and sensitivity of inputs to the engine both on the ground and in flight. Testing should determine that lost motion and friction control do not affect range or accuracy of control. Where power elements are involved in the control action, rates and sensitivity should be determined for all applicable control conditions.

VERIFICATION LESSONS LEARNED

3.5.1.1 **Control linkage adjustment.** Adjusting points shall be provided in the interface control linkages.

REQUIREMENT RATIONALE (3.5.1.1)

Adjustments in the control cable rigging may be required.

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REQUIREMENT GUIDANCE

The number of adjusting points should be kept to a minimum. Adjustment points should be at the point of mating of any two control elements and the adjustment may be in either the mating or mated element or both. A locking device should be included in each adjustment point. Linkage components which are not designed for service adjustment should be correctly adjusted at the time of initial assembly and should be secured to prevent in-service alteration.

REQUIREMENT LESSONS LEARNED

4.5.1.1 Control linkage adjustment. Interface control linkage adjustment points shall be verified by inspection.

VERIFICATION RATIONALE (4.5.1.1)

Control linkage adjustment points are readily verified by inspection.

VERIFICATION GUIDANCE

A physical inspection of linkage adjustment points should be made to assure accessibility and maintainability.

VERIFICATION LESSONS LEARNED

3.5.2 Cockpit control lever functional characteristics

3.5.2.1 Control levers in multiengine aircraft. The control levers in multiengine aircraft shall occupy the same relative quadrant position for the same control setting throughout the entire engine power range.

REQUIREMENT RATIONALE (3.5.2.1)

Sound human engineering design dictates that identical positioning of control levers for the same control setting is essential to achieve a compatible pilot/vehicle operating interface. The single cockpit control lever position for all engines readily facilitates balancing engine thrust outputs.

REQUIREMENT GUIDANCE

If automatic boost controls are used on multiengine aircraft, the controls shall have a minimum number of adjustment points to enable matching of the power levers to within 0.125 inch throughout the quadrant travel as measured along the profile of the quadrant while advancing and retarding the control.

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REQUIREMENT LESSONS LEARNED

4.5.2 Cockpit control lever functional characteristics

4.5.2.1 Control levers in multiengine aircraft. The relative quadrant position of the control levers in multiengine aircraft shall be verified by inspection.

VERIFICATION RATIONALE (4.5.2.1)

The relative quadrant position of each control lever is easily verified by inspection.

VERIFICATION GUIDANCE

Control lever(s) relative quadrant positions should be inspected during ground testing for a range of power settings.

VERIFICATION LESSONS LEARNED

3.5.2.2 Control quadrant detents and stops. The control quadrant shall have a detent at the _____ position(s). Control lever stops which require a deliberate motion to defeat shall be used at the _____ position(s).

REQUIREMENT RATIONALE (3.5.2.2)

Control quadrant detents indicate control lever position to the pilot therefore enhancing the pilot/engine interface. Control quadrant stops are necessary to prevent inadvertent and undesirable positioning of the control lever.

REQUIREMENT GUIDANCE

Power control lever detent postitions should be marked for easy location identification while operating the power control lever. Control lever position detents may be required at quadrant positions, such as FUEL ON, FUEL OFF and intermediate power (to prevent inadvertant positioning into augmented engine operation).

For turbojet and turbofan engine controls the cockpit power lever quadrant should provide an idle stop which will prevent the control from being retracted to the cutoff position without requiring a deliberate offsetting motion to defeat the stop. The control linkage should be rigged such that, upon movement of the control lever, the stops which limit the travel of the

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engine component are contacted prior to contacting the stops on the control quadrant. The control lever should be designed to contact the quadrant stop upon application of a control lever force of not more than 250 percent nor less than 200 percent of the force needed to move the engine component including overcoming the friction in the control system.

REQUIREMENT LESSONS LEARNED

4.5.2.2 Control quadrant detents and stops. Control quadrant detents and stops shall be verified by inspection and demonstration.

VERIFICATION RATIONALE (4.5.2.2)

The quadrant stops' ability to prevent an inadvertent, undesirable positioning of the control lever is most effectively and economically verified by inspection and demonstration.

VERIFICATION GUIDANCE

Quadrant detents and stops should be verified during ground testing of the cockpit controls.

VERIFICATION LESSONS LEARNED

3.5.2.3 Friction lock. An adjustable friction device shall be incorporated in the throttle control mechanism.

REQUIREMENT RATIONALE (3.5.2.3)

A friction lock is required to retain the power levers in a set position.

REQUIREMENT GUIDANCE

The friction device, if incorporated on the throttle quadrant, should permit adjustment by the pilot without requiring removal of his hand from the power lever. The direction of actuation of the device should be forward, upward, or clockwise to increase friction. A friction lock may not be required; depending upon the control system design, adequate friction may be provided. In a control system that incorporates a throttle snatcher (retarding the throttle to idle position when a thrust reverser is inadvertently deployed), the control system could be damaged by incorporating a friction lock.

REQUIREMENT LESSONS LEARNED

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4.5.2.3 Friction lock. The incorporation of an adjustable friction lock device shall be verified by inspection and demonstration.

VERIFICATION RATIONALE (4.5.2.3)

Inspection and demonstration methods are proven techniques for verifying the existence and operation of a friction lock device.

VERIFICATION GUIDANCE

Verification of the friction lock device operation should be accomplished during ground testing.

VERIFICATION LESSONS LEARNED

3.5.3 Cockpit control lever performance

3.5.3.1 Control lever forces. The maximum force required to move the cockpit _____ shall not exceed _____ pounds per engine and _____ pounds for simultaneous movement of all levers.

REQUIREMENT RATIONALE (3.5.3.1)

The maximum force required to move the cockpit engine control levers should be defined to assure compatibility with good human engineering design practice.

REQUIREMENT GUIDANCE

AFSC DH 1-3, DN 2D18 provides guidance on the design of continuous adjustment linear controls. The maximum force required to move power levers should not exceed 7.5 pounds per engine and 30 pounds for all engines. In turboprop installations, the maximum force required to actuate the turboprop condition lever should not exceed 7.5 pounds per engine. The maximum force on all other controls should not exceed 10 pounds per control. The forces specified herein should include both the force necessary to overcome friction in the entire system and the force necessary to actuate the engine element. Movement of the power lever throughout the operating range should be free of abrupt changes in actuating force. The maximum force to be specified is applicable in either direction of movement and under all conditions of operation, including static conditions with the engine not operating, and at any temperature from +200°F (+93.3°C) to -65°F (-54°C); however, the maximum breakaway force allowable after a low temperature soak should not exceed 20 pounds. Force amplifiers, whether electric, hydraulic, or pneumatic, may be used to reduce the operating force provided that upon amplifier failure the control system will continue to meet the force requirements specified.

REQUIREMENT LESSONS LEARNED

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4.5.3 Cockpit control lever performance

4.5.3.1 Cockpit control lever forces. The cockpit control lever force requirements shall be verified by test.

VERIFICATION RATIONALE (4.5.3.1)

Verification of control lever forces by test is essential to assure that a sound "human engineered" control is provided for the air vehicle crew.

VERIFICATION GUIDANCE

The forces specified in 3.5.3.1 include both the force necessary to overcome friction in the entire system and the force necessary to actuate the engine element. Control actuation forces for the augmentor thrust reverser and thrust vectoring nozzles or vanes should be measured as applicable. All forces should be measured at the center of the control knob with the friction locking device, if any, in the extreme loose position. Forces on the controls should be applied tangentially at the center of the knob. The maximum force specified is applicable in either direction of movement and under all conditions of operation, including static conditions with the engine not operating, and at any temperature from +200°F (+93.3°C) to -65°F (-54°C); however, the maximum breakaway force allowable after a low temperature soak should not exceed 20 pounds.

VERIFICATION LESSONS LEARNED

3.5.3.2 Control lever deflection. The deflection of the _____ lever(s) while in the operating range shall not exceed _____ degrees in each direction when held rigid at any intermediate position and a force of _____ applied.

REQUIREMENT RATIONALE (3.5.3.2)

The capability to accurately position the cockpit power control levers is necessary for precise and repeatable engine control.

REQUIREMENT GUIDANCE

Throttle and propeller control systems which do not include means for amplifying the applied force should be designed such that the movement of each lever while in the operating range should not exceed 2 degrees in each direction when the control linkage is held rigid at any intermediate position and when a force of 150 percent of the force required to move the control is applied. In controls which include force amplifiers, the deflection of the control system should be defined as the amount that the control knob will lead the theoretical position (no deflection) while developing 60 pound-inches of output torque. This deflection should not exceed 2 degrees in each direction.

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Cockpit controls which are hand-actuated by push-pull motions should not move in excess of 0.062 inch from center position in either direction when the control is held rigid at the controlled device and a force of 150 percent of the force required to move the control is applied. The force on the controls should be applied tangentially at the center of the knob for quadrant controls and in the direction of motion for push-pull controls. The deflections specified above are the maximum allowable on multiengine aircraft and should be materially reduced for single-engine aircraft where the control can be made more direct. Application of the maximum allowable control lever forces should not subject the controlled component to loads in excess of those specified for the component.

REQUIREMENT LESSONS LEARNED

4.5.3.2 Control lever deflection. Control lever deflection shall be measured with the specified force applied to verify compliance with 3.5.3.2.

VERIFICATION RATIONALE (4.5.3.2)

Control lever deflection should be measured to assure that the lever can be precisely controlled and positioned.

VERIFICATION GUIDANCE

The movement of each lever while in the operating range should not exceed the specified number of degrees in each direction when the control linkage is held rigid at any intermediate position and the specified force required to move the control is applied. In control systems which include force amplifiers, the deflection of the control system is the amount that the control knob will lead the theoretical position (no deflection) in either direction while developing a specified output torque.

VERIFICATION LESSONS LEARNED

3.5.4 Turbojet and turbofan cockpit controls. A single cockpit power lever for control of each engine shall be installed in turbojet and turbofan powered aircraft which coordinates all engine variables through a single engine connection. The power lever quadrant shall be configured in accordance with figure 1, item A. Power lever angular travel shall _____.

REQUIREMENT RATIONALE (3.5.4)

A cockpit power lever for each engine facilitates individual engine control if necessary (i.e., during emergencies and ground maintenance).

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REQUIREMENT GUIDANCE

Power lever angular travel should not exceed 110 degrees ± 10 degrees. The control lever should be capable of unrestricted advancement from the cutoff position to any forward position.

REQUIREMENT LESSONS LEARNED

4.5.4 Turbojet and turbofan cockpit controls. A single cockpit power lever which coordinates all engine variables through a single engine connection shall be verified by inspection. Power lever angular travel shall be verified by test.

VERIFICATION RATIONALE (4.5.4)

Verification of the requirements of 3.5.4 by inspection and test have proven to be the most economical and effective methods available to verify this requirement.

VERIFICATION GUIDANCE

An inspection of the cockpit power lever and measurement of the power lever angular travel should be made during ground testing of the propulsion system.

VERIFICATION LESSONS LEARNED

3.5.4.1 Thrust reverser controls. A power lever quadrant in accordance with figure 1, item B, shall be provided for engine installations which incorporate thrust reversers. Thrust reverser controls shall be as follows: _____.

REQUIREMENT RATIONALE (3.5.4.1)

A standard quadrant design is desired for all Air Force aircraft.

REQUIREMENT GUIDANCE

A lift gate should be used to permit entrance to the reverse thrust position. A mechanical interconnect should be included to prevent increasing engine power above idle thrust until the thrust reverser is completely extended. Further travel in the aft direction should increase reverse thrust by increasing engine power. The full reverse throttle stop should be adjustable. Control advancement from the reverse thrust position should be restricted to idle thrust until the thrust reverser is completely retracted.

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REQUIREMENT LESSONS LEARNED

4.5.4.1 Thrust reverser controls. Thrust reverser controls shall be verified by inspection and demonstration.

VERIFICATION RATIONALE (4.5.4.1)

The functional characteristics and quadrant configuration of the thrust reverser controls have typically been verified by demonstration and inspection.

VERIFICATION GUIDANCE

The quadrant configuration should be inspected prior to the start of ground testing and demonstrated during ground and, if appropriate, flight testing.

VERIFICATION LESSONS LEARNED

3.5.5 Turboprop cockpit controls. Two flight station controls for each engine-propeller combination shall be installed in turboprop aircraft; one engine power lever and one turboprop condition lever shall be provided. These controls shall have functional characteristics substantially in accordance with figure 1, items C and D.

REQUIREMENT RATIONALE (3.5.5)

A standard turboprop cockpit control design is desired for all Air Force aircraft.

REQUIREMENT GUIDANCE

The functional characteristics of the cockpit controls should be designed in accordance with figure 1, items C and D.

REQUIREMENT LESSONS LEARNED

4.5.5 Turboprop cockpit controls. Flight station controls and their functional characteristics shall be verified by inspection and demonstration.

VERIFICATION RATIONALE (4.5.5)

Flight station controls and their functional characteristics are easily verified by inspection and demonstration.

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VERIFICATION GUIDANCE

Turboprop engine controls should be verified by inspection and demonstration during propulsion system ground testing.

VERIFICATION LESSONS LEARNED

3.5.5.1 Turboprop engine power lever. The power lever shall be in accordance with the standards shown on figure 2. In multiengine aircraft the power lever shall have a minimum vertical lift of _____. In single console installations (one set of power levers for both pilots) the maximum upward pull force shall not exceed _____ per lever. In dual console installations (one set of power levers for the pilot and a connected set for the copilot) the maximum upward pull force shall not exceed _____ per lever. The vertical lift line shall be at _____ to the plane of the pilot's seat. The upper _____ below the knob of each power lever shall be spaced as shown in the lower view on figure 2. The levers within this area shall have the rounded edges and a thickness of _____.

REQUIREMENT RATIONALE (3.5.5.1)

A standard turboprop power lever design is desired for all Air Force turboprop aircraft.

REQUIREMENT GUIDANCE

In multiengine aircraft the power lever should have a minimum vertical lift of 1.25 inches. In single console installations (one set of power levers for both pilots) the maximum upward pull force should not exceed 1 pound per lever. In dual console installations (one set of power levers for the pilot and a connected set for the copilot) the maximum upward pull force should not exceed 3 pounds per lever. The vertical lift line should be at 90 degrees $\pm$10 degrees (maximum) to the plane of the pilot's seat. The upper 2 inches below the knob of each power lever should be spaced as shown in the lower view on figure 2. The levers within this area should have the rounded edges and a thickness of 0.125 inch

REQUIREMENT LESSONS LEARNED

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4.5.5.1 Turboprop engine power lever. The physical configuration and functional force limitations shall be verified by inspection and test.

VERIFICATION RATIONALE (4.5.5.1)

The verification methods of inspection and test have been proven by experience to be the appropriate means of verifying power lever characteristics.

VERIFICATION GUIDANCE

The physical configuration of the power lever should be verified by inspection and measurement. Upward pull forces should also be measured.

VERIFICATION LESSONS LEARNED

3.5.5.2 Turboprop condition lever. The condition lever shall be in accordance with the standards shown on figure 3.

REQUIREMENT RATIONALE (3.5.5.2)

A standard turboprop condition lever is desired for all Air Force aircraft.

REQUIREMENT GUIDANCE

The turboprop condition lever should be designed in accordance with figure 3.

REQUIREMENT LESSONS LEARNED

4.5.5.2 Turboprop condition lever. The physical dimensions and functional characteristics of the condition lever shall be verified by inspection and test.

VERIFICATION RATIONALE (4.5.5.2)

The verification methods of inspection and test have been proven by experience to be the appropriate means of verifying condition lever characteristics.

VERIFICATION GUIDANCE

The physical configuration of the condition lever should be verified by inspection and measurement.

VERIFICATION LESSONS LEARNED

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3.5.5.3 Reverse-pitch propeller controls. When reverse pitch propellers are installed, reverse pitch shall be obtained by movement of the engine power lever past the idle position in the aft direction through the lift point. Motion of the power lever shall energize the reversing circuits of the propeller control. Further travel in the aft direction shall increase reverse thrust by increasing engine power. The quadrant shall include a positive stop to prevent inadvertent reverse-pitch operation regardless of the landing gear position. The stop shall consist of an enclosed track having a vertical lift at the flight idle position and shall be bypassed by vertically raising the power lever. Movement of the lever into the aft or reverse position shall not require removal of the operator's hand from the lever. Forward motion of the control shall be unrestricted and shall unreverse the propeller. In multi-engine aircraft equipped with two sets of controls, the power lever shall permit lift-to-reverse operation from either pilot's control. The controls shall permit reversal of any propeller independently of the others.

REQUIREMENT RATIONALE (3.5.5.3)

A standard reverse-pitch propeller control is desired for all Air Force turboprop aircraft.

REQUIREMENT GUIDANCE

The reverse-pitch propeller control should be designed in accordance with the requirements above.

REQUIREMENT LESSONS LEARNED

4.5.5.3 Reverse-pitch propeller controls. The design, operation and function of the reverse-pitch propeller controls shall be verified by demonstration and test.

VERIFICATION RATIONALE (4.5.5.3)

Experience has shown that demonstration and test are the preferred methods for verifying reverse-pitch propeller controls.

VERIFICATION GUIDANCE

Reverse-pitch propeller controls should be demonstrated and tested during propulsion system ground testing.

VERIFICATION LESSONS LEARNED

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3.6 Quick engine change assembly. The quick engine change (QEC) assembly shall consist of the completely assembled propulsion system package with all components installed which can be readily transported for installation or removal from the aircraft. Components of the QEC, which include the engine and those equipments and accessories attached to or which form a part of the engine and its installation, shall be defined.

REQUIREMENT RATIONALE (3.6)

A definition of the QEC is required to establish a mutual language and define a design baseline.

REQUIREMENT GUIDANCE

The design of the QEC assembly should consider ease of assembly/disassembly, interchangeability, disconnect location, type and number of connections required for mating the airframe and engine systems. The QEC design should be compatible with standard handling equipment and tools. QEC module components should be designed with a means for handling during assembly or disassembly. Ground handling mounts, which should be readily accessible without removal of engine components, engine, all engine mounted equipment and accessories and operating fluids. The handling mounts should withstand loads of 4 g axial, 2 g lateral, and 3 g vertical.

REQUIREMENT LESSONS LEARNED

4.6 Quick engine change assembly. The quick engine change (QEC) assembly shall be inspected and the equipment and accessories which compose the QEC verified.

VERIFICATION RATIONALE (4.6)

The QEC is easily verified by inspection.

VERIFICATION GUIDANCE

The QEC should be verified by witnessing the build-up of an assembly.

VERIFICATION LESSONS LEARNED

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3.6.1 Interchangeability. Multiengine aircraft QECs shall be completely interchangeable between positions without physical or electrical modifications to any part of the QECs and without resorting to QEC sectioning.

REQUIREMENT RATIONALE (3.6.1)

Interchangeability enhances propulsion system supportability and improves maintainability.

REQUIREMENT GUIDANCE

Where possible, design the interchangeability so that all parts, assemblies and subassemblies, units, etc. have the same manufacturer's part number and are directly and completely interchangeable.

REQUIREMENT LESSONS LEARNED

4.6.1 Interchangeability. Interchangeability of the QEC between the several locations on the aircraft shall be demonstrated, including determination of the normal time, personnel and support requirements for removal and complete reinstallation.

VERIFICATION RATIONALE (4.6.1)

Experience has shown that a demonstration of interchangeability is necessary to verify this requirement.

VERIFICATION GUIDANCE

The demonstration outlined should be performed by an experienced crew using standard tools. A record of the number of personnel, manhours and elapsed time to complete each operation should be kept. Such a record should include the time necessary to position hoists and slings, disconnect and reconnect lines and electrical circuits, drain and refill oil and fuel systems as necessary, and time for cowl removal and reinstallation. A record of any items removed from the engine or QEC should be made. The tool and facility requirements, the acceptability of each of the operations involved in removal and replacement, and any conditions which are sensitive or potential sources of trouble in the removal and replacement operation should be noted.

VERIFICATION LESSONS LEARNED

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3.6.2 Removal and installation. The time required to remove a QEC in operating condition and to install another QEC and put it in operating condition shall not exceed _____ minutes.

REQUIREMENT RATIONALE (3.6.2)

The capability to rapidly change a QEC enhances aircraft maintainability and mission readiness.

REQUIREMENT GUIDANCE

The time required to remove a QEC in operating condition and to install another QEC and put it in operating conditon should not exceed 30 minutes. Cowlings should be provided in order to permit easy access to and rapid removal of the QEC. When practicable, cowlings should be hinged to permit rapid access to the QEC. Where appropriate, cowling sections may also serve as workstands for maintenance personnel.

A QEC exchange should be accomplished with standard tools. Application of unique tools should be minimized and avoided if possible. The QEC should be installed such that it can be readily removed as a unit without detaching any basic engine components or accessories directly mounted on the engine or without removing any significant parts of the aircraft structure. In turboprop installations, if the propeller is not part of the QEC, the elapsed time should include removal of the propeller from the installed engine and installation of the same propeller on the replacement engine.

Quick-disconnect mechanical and electrical couplings should be used at all connections between the QEC and the airframe. The disconnects should be arranged to preclude inadvertent reversal, mismatch, or improper connection of electrical and mechanical connectors, couplings, and fittings. Disconnects should be located in convenient groupings and should be arranged so that several mechanics may simultaneously work in the area with a minimum of interference. Disconnects for QECs should be marked with a solid band of international orange paint conforming to TT-P-59.

Quick-disconnect fluid line couplings should be in accordance with MIL-C-7413 and MIL-C-25427. All fluid lines passing through firewalls should have flared tube bulkhead fittings or equivalent. Fluid couplings of the quick disconnect type which are used at firewalls should have a bulkhead fitting end for installation at the firewall.

The QEC and its modules should be designed for installation/removal and assembly/disassembly in a nuclear, biological and chemical environment.

REQUIREMENT LESSONS LEARNED

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4.6.2 Removal and installation. A QEC in operating condition shall be removed and replaced to an operating condition in _____ minutes.

VERIFICATION RATIONALE (4.6.2)

Demonstration of a 30-minute removal and replacement capability has proven to be the only indisputable means of demonstrating compliance with this requirement.

VERIFICATION GUIDANCE

The demonstration outlined should be performed by an experienced crew using standard tools. A record of the number of personnel, manhours and elapsed time to complete each operation should be kept. Such a record should include the time necessary to position hoists and slings, disconnect and reconnect lines and electrical circuits, drain and refill oil and fuel systems as necessary, and for cowl removal and reinstallation. A record of any items removed from the engine or QEC should be made. The tool and facility requirements, the acceptability of each of the operations involved in removal and replacement, and any conditions which are sensitive or potential sources of trouble in the removal and replacement operation should be noted.

VERIFICATION LESSONS LEARNED

3.7 Fire protection. The engine installation shall be designed and protected from the hazards of fire in accordance with _____.

REQUIREMENT RATIONALE (3.7)

The engine installation should be designed and protected from fire hazards to maximize safety of the air vehicle, crew, and ground personnel.

REQUIREMENT GUIDANCE

The engine installation should be designed and protected from the hazards of fire in accordance with the requirement tailored from MIL-F-87168. Fire detection systems should be installed in all zones containing combustible fluids and ignition sources and in which a fire or explosion could occur as a result of failure of any system or equipment within that zone. An overheat detection system should be installed in all zones where structural components or equipment can be overheated by abnormal engine operation or other failures and where an overheat condition will present a flight hazard or add to an existing hazard. Fire extinguishing systems should be installed on all multiengine aircraft. The system should provide protection for all potential fire zones. An extinguishing system should be installed on experimental single engine aircraft where the complexity of the engine installation, the inclusion of new and unproven features, or the general importance of the project warrants additional protection for the aircraft. Rotary wing and hover craft with a single engine should require a one-shot fire extinguishing

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system. A ground fire access door having minimum dimensions of 5.5 by 10 inches should be provided for each fire zone. This door should be in such a position that the nozzle of a ground-type extinguisher can be thrust against the door, forcing it open and permitting the extinguishing agent to be injected directly into the compartment. The retaining device should be sufficiently strong to hold the door shut against airloads. Quick-release latches should not be used. The door should be accessible to ground personnel and should be located where burning flammable fluid will not drain on the operator of the extinguisher nozzle. The door should be marked ACCESS FOR FIRE EXTINGUISHER. All fuel tanks should be separated from the engine nacelles, from the interior of the wing, and from the interior of the fuselage with a liquidproof and vaporproof barrier. The engine installation of all aircraft, regardless of the number or relative position of the engines, should incorporate a liquid and vapor barrier which separates the accessory and compressor sections or that section of the compartment which contains the greater part of the combustible fluid lines and accessories from any engine surface which exceeds a temperature of 700°F. The barrier should prevent combustible fluid and vapor contact with the hot sections of the engine. If it is not practical to isolate engine surfaces which exceed temperatures of 700°F, the accessories and drives should be remotely located and isolated from the engine compartment. All fire zones and engines should be separated from one another, from fuel tanks, and from the interior of the wings and fuselage by a fireproof firewall. All firewalls should be constructed of 0.015 inch thick (minimum) stainless steel or other material which will withstand a 2,000°F flame for 15 minutes without burn through in the presence of airflow. The firewalls should be liquidproof and vaporproof. Materials used on each face of firewalls, or immediately adjacent thereto, should be of a type that will not ignite as a result of heat transfer from a flame on the opposite side of the firewall. Adequate spacing should be maintained between the engine and the firewall in order that a flame from a ruptured burner will not burn through the firewall. All flammable fluid lines and hose assemblies located in any potential fire zone and engine compartment bleed air lines should be constructed of stainless steel or other material having equivalent fire resistance; the lines and hose assemblies should withstand a 2,000°F flame for at least 5 minutes without leakage. The fire resistant capability should be maintained when the lines and hose assemblies are exposed to the minimum fluid flow rates, the maximum fluid temperatures, and the normal vibrational environments. Fittings and supports should have an equal resistance to fire. This guidance should not apply to vent and drain lines which upon failure will not contribute to any existing fire hazard. All equipment and components which contain flammables and are located in a fire zone should meet the same fire resistant criteria. The use of hose assemblies should be held to a minimum in order to reduce the volume of flammable fluid available in a fire situation. Combustible fluid lines, including vent lines, should not penetrate any nacelle or engine compartment except the ones they feed. In no case should combustible fluid lines be routed through air inlet ducts or plenums. No equipment, components, or lines should be permitted in a fire zone unless they are required for the operation or functioning of equipment contained in the fire zone. The surface temperature of all bleed air lines or ducts located in any fire zone should not exceed a temperature of 700°F under any operating condition.

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Firewall shutoff valves should be used in all pressurized lines carrying combustible fluids to any fire zone. Shutoff valves need not be provided for drain lines and lines used with a closed loop fluid system peculiar to a system or equipment within the fire zone; e.g., the integrated drive generator oil cooling system lines between the generator and the heat exchanger. The shutoff valve should be located as near the fire zone as practicable, but not in the zone. If possible, the valve should be located on the outside face of the firewall which isolates the fire zone. Electrically operated valves should not change position during flight as a result of a short circuit, stray voltage, electrical power failure, or normal mechanical forces.

REQUIREMENT LESSONS LEARNED

4.7 Fire protection. The fire protection system shall be verified in accordance with the applicable verification requirements of _____.

VERIFICATION RATIONALE (4.7)

Experience has shown that a high level of confidence in the capabilties of the fire protection system is best obtained by accomplishing the proven verification procedures of MIL-F-87168.

VERIFICATION GUIDANCE

The fire protection system should be verified in accordance with the verification requirements tailored from MIL-F-87168. As a minimum, an analysis of material and component tests should be conducted to confirm that the requirements of MIL-F-87168 regarding liquid and vapor barriers, firewalls, equipment and hardware in potential fire zones, and fluid shutoff valves have been met. A 24 inch by 24 inch specimen of each type of firewall employed in the propulsion installation system should be tested in accordance with the "Fireproof Tests" specified in ARP 1055. The burner should produce a flame temperature of 2,000°F measured 0.25 inch in front of the test specimen and have the correct heat (BTU) release. Representative samples of tubes, hose assemblies, fittings, supports, and other related components which carry flammable fluids and are located in potential fire zones should be subjected to a fire resistance test. The test should be as specified in the "Fire Resistance Tests" of ARP 1055 except that the fluid flow rate should be the lowest rate expected and the liquid temperature should be the maximum expected to be encountered during aircraft operation. Ground testing of the fire and overheat detection system and the fire extinguishing system should also be accomplished.

VERIFICATION LESSONS LEARNED

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3.8 Drains. Closed compartments and equipment subject to fluid spillage or leakage shall be provided with drains for routing the fluid leakage or spillage overboard or to a collector system. The collector system shall have a capacity of _____.

REQUIREMENT RATIONALE (3.8)

Drainage is necessary to prevent the hazardous accummulation of liquids.

REQUIREMENT GUIDANCE

All closed compartments in the engine installation or nacelle and pylon such as the engine accessory section, spaces enclosing fuel, oil and hydraulic lines and equipment, vent areas and other pockets where fluids may collect should have suitable drainage provisions for all normal ground and flight attitudes. All drains should be identified with labels or other markings. Inadvertent liquid spillage and accidents as well as combat aircraft battle damage should be considered when sizing and locating drains. The engine installation should be provided with a drainage system to route collected fluids overboard free of the aircraft or to an accessible central collection point for on the ground maintenance removal. Overboard drain lines should be routed to permit fluid to exit free of the aircraft fuselage, nacelle, wing and pylon and should pass through resilient grommets in bulkheads and cowlings in order to prevent chafing. Drain masts should be compatible with the most adverse fluid they may come in contact with externally. Drain exits from the aircraft should be scarfed in the air stream direction so that a scavenging suction effect is produced in flight by airflow around the drain discharge opening assuring positive drainage of all fluids. The drain opening should be located in an area where the local airstreams are such that drained fluids and vapors will exit freely and will not be driven back into the drained compartment or any other compartment or the engine exhaust gas wake. Central collection points for on the ground fluid removal may include devices such as ecology kits or tanks to collect drained fluids for protection of the environment. Ecology kits and tanks should have the capability to retain fluids without overflow under maximum normal drainage rates. Drain holes should be provided in pylons, bulkheads, stiffeners, and the skin in order to permit the normal flow of accumulated fluids to collect at low points. Drains may be interconnected if line sizes are adequate to ensure proper drainage. Interconnection should not be permitted where return of any fluid or vapor may create a fire hazard or damage any of the components whose drains are interconnected. When a single drain opening is used for the entire engine accessory section drain system, it should be sized to accommodate all leakage. Individual drain holes and drain tubing should be sized with an internal diameter that will prevent clogging due to dirt, insects, or debris. The drain system should not contain traps where condensation and fluids can accumulate, freeze, or cause corrosion. Drainline materials should be compatible with the drained fluids and the compartment environment. Lines connected to structural bulkheads should be of flex lines and fittings. Electrical component drains should not be manifolded with fluid system drains. Fluid drains of combustion areas should be such that excess combustible fluids will not collect in these areas after engine shutdown or false starts with the engine in a level position, a nose up position or nose down position. Drainage of engine afterburner ducts should be provided to prevent accumulation of fuel in the duct while the engine is in its normal non-operating position.

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REQUIREMENT LESSONS LEARNED

4.8 Drains. Analyses shall be completed to determine that drain line sizes, pressure gradients, and flow conditions are compatible for proper drain system function relative to the propulsion equipment and installation. Propulsion installation system drains which, due to their configuration or location, type and quantity of fluid drained, etc., appear questionable, insofar as adequate drainage is concerned, shall be subjected to drainage tests. The drainage tests shall be conducted with dyed fluids and shall verify that the performance of all questionable and potentially hazardous drains is acceptable at all applicable aircraft ground and in flight operating attitudes and conditions. The capacity of the collection system shall be measured.

VERIFICATION RATIONALE (4.8)

A high level of confidence in the functional capabilities of the drain system is best obtained by analysis and test.

VERIFICATION GUIDANCE

Drain system operation should be observed and recorded during all ground test operation and following flight operations. An accounting should be made of all test incidents relative to the drain system such as stopped or sluggish drains, excessive backpressure, failures, contamination, fire hazards, areas omitted that should have drains, or other conditions not in accordance with recommended design practice. The required dyed fluid tests should be conducted early in the test program, consistent with safety requirements and identification of drains requiring the tests.

VERIFICATION LESSONS LEARNED

3.9 Plumbing and wiring. Plumbing and wiring shall be installed in accordance with _____.

REQUIREMENT RATIONALE (3.9)

The engine installation plumbing and wiring is critical to reliable and safe air vehicle and engine operation and should therefore be installed in accordance with established and proven design practice.

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REQUIREMENT GUIDANCE

Hydraulic and fuel system plumbing and electrical wiring installations should conform to specifications tailored from MIL-H-87227, MIL-F-87154 and MIL-E-87219, respectively. All piping should be routed as directly as practicable, considering vulnerability, accessibility, and compartment obstruction. The positioning of combustible fluid lines relative to engine components and engine bleed air ducts should be accomplished with the intent of minimizing the fire hazard associated with fluid leakage through accident and combat damage. Plumbing and wiring should be routed such that no lines pass unnecessarily close to high temperature components. Fuel, oil, and hydraulic lines and equipment should be located to minimize the possibility of leaking fluid coming into contact with electrical equipment through either the effect of gravity, airflow, or combat damage. All combustible fluid lines should be routed below electrical equipment and wires. All fluid lines should be routed to minimize interference with removal or adjustment of engine accessories and in the most efficient manner to obtain maximum protection from the aircraft structure against combat damage. Where practicable, fuel system plumbing should be routed through fuel tanks in order to minimize fuel losses and fire hazards resulting from combat damage or fitting failure. Combustible fluid lines should not be used to support other components, plumbing, wiring, etc. The components and systems should be provided with integral or built-in fluid lines when practicable. Fluid lines should be free from sharp bends and restrictions. Access for adequate maintenance and inspection of lines and fittings should be provided without necessitating engine removal.

Tubing should be fabricated and installed such that forcing, bending, or stretching is not necessary to accomplish installation. No tube assembly should be installed or secured in a stressed condition. Metal tubing should be installed such that vibration or deflection of the aircraft structure to which it is attached will not cause failure of the tubing or any of its connections and supports. Metal tubing should be supported in accordance with MIL-H-87227 and MIL-F-87154 and should comply with the minimum bend radii specified in MS33611. Flexible connections should be used on all fluid lines from the aircraft structure to the engine. Adequate spacing should be maintained between all tubing and between the tubing and the aircraft structure to prevent chafing during any static and flight conditions and as a result of system pressure pulses. All tubing should be protected against chafing by the use of resilient grommets or protective coverings in those areas having inadequate clearance. Where clamps or blocks are used for tubing support, resilient material should be used between the clamp or block and the tube. The resilient material should be compatible with the environment, the effects of temperature and exposure to fuel, oil, and hydraulic fluid should not cause material deterioration.

Hose assemblies should be in accordance with MIL-H-8795, MIL-H-25579, or MIL-H-38360. Installed hose assemblies should meet the following criteria:

a. Hose assemblies should consist of hose and end fittings in accordance with requirements of applicable specifications and standards. Hose clamp beaded tube type installations should not be permitted.

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b. Hose installations should be designed so that normal motion does not impose torsional deflection under any condition of operation which will loosen the hose connecting fittings.

c. Hose assemblies carrying flammable fluids in a potential fire zone should meet the applicable requirements of MIL-F-87168.

d. Unless otherwise specified in the hose specification, hose installation and operational bend radii should conform to MS33790.

e. Excessive motion should be restrained where necessary to prevent chafing or misalignment; however, the hose should not be rigidly supported except at the end fittings.

f. Hose assemblies should be selected and installed so that elongation or contraction under pressure, within the hose specification limits, will not be detrimental to the installation, either by causing added strains on the end fitting or by binding or chafing of the hose.

g. Hose assemblies should have an appropriate length tolerance. The tolerance, both maximum and minimum, should be considered and designed into the installation.

h. Hose assemblies which have permanent fittings should be installed with adequate installation and wrenching clearances to permit replacement with a hose assembly which has replaceable fittings.

i. Hose assemblies should be supported in accordance with MIL-H-87227 and MIL-F-87154.

Permanent line joints should be used at all points where line separation is not required to facilitate production or to permit normal aircraft servicing and simplify maintenance; however, all lines should have a minimum number of fittings. Fittings conforming to MS or AN standards should be used where suitable. Where military standards are not suitable, separable line joints should meet the following requirements:

a. The joint should be leakproof throughout the environmental range of its application.

b. The joint should accommodate practical variations in material and handling techniques encountered in service.

c. The joint should permit repeated assembly and disassembly without attrition of components.

d. Joint effectiveness should not deteriorate during the useful life of the airframe.

e. The joint design should minimize the probability of improper joint assembly or installation.

f. The design should be as simple as possible.

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g. The joint should permit maximum ease of installation and removal without use of special tools. (Note: Tools which are listed in federal stock lists, such as AN standard spanner wrenches, strap wrenches, torque wrenches, and deep sockets, are not to be considered special tools.)

h. The joint should not require critical torque techniques for proper installation.

i. The joint should be compatible with the plumbing member to which it is attached.

j. The joint should not be sensitive to cold flow, swell, or shrinkage of synthetic components.

k. The joint should not be subject to torque relaxation.

l. The joint should accommodate established structural requirements plus any self-imposed loads while maintaining those safety factors presently imposed on existing standard fittings.

REQUIREMENT LESSONS LEARNED

4.9 Plumbing and wiring. The plumbing and wiring installation shall be verified in accordance with the requirements of _____.

VERIFICATION RATIONALE (4.9)

Verification methods that have proven to be fair and reliable through repeated use and experience are preferred.

VERIFICATION GUIDANCE

Hydraulic and fuel system plumbing and electrical wiring installations should be verified in accordance with the verification requirements tailored from MIL-H-87227 and MIL-F-87154. As a minimum, the following inspections and tests should be completed:

a. Prior to the commencement of any testing of the propulsion system, all plumbing and wiring within the engine bay or nacelle and pylon should be inspected.

b. Ground starts and engine operation combined with various system functions should be conducted to subject the plumbing and wiring installation to the range of environmental conditions during operation. Flight tests should be conducted to achieve the full range of conditions.

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This testing should determine that the plumbing and wiring system exhibits no destructive vibration characteristics, operates within permissible temperatures, does not fail or exhibit objectionable inspection characteristics, and meets all requirements of the aircraft.

After the completion of each series of tests on the propulsion system, all plumbing and wiring installations within the engine/QEC and pylon should be inspected. Unless otherwise specified by the procuring activity, these inspections should be conducted at 50-hour intervals during the flight test program. When appropriate, additional inspections should be conducted after the following flight conditions and maneuvers, as applicable:

a. Simulated and actual weapon delivery

b. Internal gun firing

c. Inflight refueling

d. Maximum positive and negative g maneuvers

e. Maximum roll and sideslip maneuvers

f. Maximum climbs and descents

g. Critical speed brake or thrust reverser actuations

Inspections of the plumbing and wiring installation (fuel, hydraulic, bleed air, drain, and vent lines; wiring bundles; control cabling; etc) should be made to determine the following:

a. Evidence of the lack of structural adequacy of the tubing, cabling, and wire bundle mounting provisions.

b. Evidence of tubing, cabling, and wiring bundle deflection, interference, and resonance.

c. Evidence of connector and fitting looseness.

VERIFICATION LESSONS LEARNED

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3.10 Structural provisions. The propulsion system installation shall be designed in accordance with the applicable structural design requirements of _____.

REQUIREMENT RATIONALE (3.10)

Components of the propulsion system installation such as the inlet duct system, exhaust system and airframe engine mounting structure must be properly designed for all anticipated load conditions. Improperly designed structure may have detrimental affects on engine function and operation and may pose a hazard to the air vehicle and crew.

REQUIREMENT GUIDANCE

The propulsion system installation should be designed in accordance with the applicable structural design specification as tailored for the program from MIL-A-87221 and as required by MIL-STD-1530. As a minimum, components of the propulsion system installation should meet the following design requirements:

a. The inlet ducting should be of sufficient strength to withstand all temperature and pressure differentials encountered under any normal flight or engine condition as well as engine stalls. The inlet should also withstand vibrations and structural loads produced by airflow variations and those produced by the engine, thrust reverser, airframe, and armament systems.

b. Engine cowlings should be designed such that in the event of a high-pressure bleed air line rupture, the cowlings will not separate from the nacelle nor should they experience structural damage which would impair safe operation of the aircraft.

c. The structural design of the exhaust system should be capable of withstanding all loadings imparted by special equipment, such as thrust reversers and thrust vectors, as well as engine anomalies, such as late augmentor ignition.

d. The airframe engine mounting system should be capable of:

(1) Sustaining without failure or permanent deformation the engine and accessory weight under inertia, gyroscopic and torque loads likely to be encountered during and after any mode of aircraft operation or maneuver within the aircraft flight attitude and ground operational limits specified in the air vehicle specification.

(2) Maintaining the engine location and alignment under linear accelerations, angular accelerations and velocities, and engine torque while providing clearance between the engine and airframe adequate to prevent interference.

(3) Sustaining, without failure, engine loadings due to rotor seizure, rotor wheel hub separation and ditching, or ultimate acceleration loadings.

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(4) Preventing excessive engine loading from engine thermal expansion and airframe deflection.

REQUIREMENT LESSONS LEARNED

4.10 Structural provisions. Structural design requirements applicable to the propulsion system installation shall be verified by the methods specified in _____.

VERIFICATION RATIONALE (4.10)

The verification methods tailored from MIL-A-87221 have been evolved through repeated applications and are judged to be the most appropriate methods available.

VERIFICATION GUIDANCE

The verification requirements tailored from MIL-A-87221 should be applied. To verify analysis critical propulsion installation structures should be instrumented and the interface loads measured under critical flight and ground load conditions. After each engine oriented critical load condition occurring during flight and ground test, a visual inspection should be accomplished to insure that no deleterious physical interference between engine/engine components and airframe is evident and that no permanent deformations exist.

VERIFICATION LESSONS LEARNED

3.10.1 Vibratory stress. The installed propulsion system shall be free from destructive vibratory stress and wear.

REQUIREMENT RATIONALE (3.10.1)

The installed propulsion system may be susceptible to vibratory stresses which could cause rapid wear out and failures.

REQUIREMENT GUIDANCE

The installed propulsion system configuration should be analyzed and the design optimized to avoid configurations with critical frequencies in the normal operating range of the engine.

REQUIREMENT LESSONS LEARNED

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4.10.1 Vibratory stress. Stress surveys shall be conducted, as specified for the weapon system, to determine freedom from destructive stress or vibration in components of the installed propulsion system.

VERIFICATION RATIONALE (4.10.1)

Stress surveys have proven to be the most effective means of assessing the vibratory stress characteristics of the propulsion system installation.

VERIFICATION GUIDANCE

All items of the propulsion system should be installed in their flight configuration except as required for instrumentation of the equipment. Control systems should be operating to their established schedules and fully operational. The range of test conditions should be such that the complete ground and flight envelope is explored. Sufficient testing should be run at the same conditions used in the test cell so that a comparison can be made between the two sets of vibratory stress data. Tests involving crosswind conditions should be accomplished with a wind velocity of not less than 15 knots.

Vibratory stress data should be recorded for engine components, such as engine fan, compressor, turbines, reduction and accessory gearboxes, and critical static structure; propellers/rotor/fans; and other propulsion system components as applicable. Sufficient test data should be obtained to determine that all components of the propulsion system are free of destructive stresses or vibratory conditions and to define the operating conditions which result in high stresses. Sufficient supporting data should be recorded so that a complete analysis of the strain data is facilitated.

A ground vibratory stress survey should be conducted to establish engine fan, compressor, accessory gearbox, and critical static structure stress characteristics when operating in the installed environment. Propeller vibratory stress surveys and control response tests should be conducted when applicable. Testing should include steady-state and transient runs under all conditions of ground and taxi operation. The effects of the following conditions, if applicable, should be investigated: thrust reverser operation, ground nacelle cooling, crosswind operation, nacelle location on the aircraft, and any other mode of operation peculiar to the particular system which could influence engine stress characteristics.

A flight vibratory stress survey should be conducted to determine critical engine component stress characteristics, with particular emphasis on the fan and compressor, when operating in the aircraft flight environment. Propeller stress surveys and control response tests should be conducted when appropriate. The flight survey should be conducted at all conditions, for both the engine and airplane within the aircraft flight operating envelope. Surveys should include takeoff, transition, climb, descent, landing, altitude restarts, maximum yaw, and normal flight maneuvers. The investigation, where applicable, should explore engine stress conditions during combat maneuvers consistent with the mission of the aircraft, ordnance firing, thrust reverser operation, augmentation, and during any other unusual maneuver or mode of operation peculiar to a particular aircraft system which could have an effect on engine dynamic vibratory characteristics.

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VERIFICATION LESSONS LEARNED

3.11 Accessibility/maintainability. All equipment in the engine installation shall be located and mounted to permit access for rapid equipment testing, servicing, removal, inspection, and replacement using standard tools. The maintainability requirements of the engine installation designs and systems shall be compatible with overall aircraft maintainability requirements. The installation designs and systems shall give a maintenance manhour per flight hour (MMH/FH) no greater than _____.

REQUIREMENT RATIONALE (3.11)

Accessibility of the installed propulsion equipment enhances visual inspection, servicing and cleaning, checking and correcting control adjustments, ground testing, engine removal, and parts replacement.

REQUIREMENT GUIDANCE

The effort required to provide accessibility should be commensurate with the relative frequency and manhours of inspection, servicing, and repair required on the equipment. Consideration should be given to accessibility/maintainability in a nuclear, biological, chemical (NBC) environment while wearing protective clothing. The arrangment of equipment and components should be such that maintenance personnel can accomplish their work without assuming awkward positions or requiring excessive movement from position to position and such that fluids and dirt are unlikely to drop on personnel and on other components during maintenance. Assemblies which are subject to frequent removal should be attached by bolts rather than by permanent attachments using rivets. Where practicable, quick-disconnect features should be used. Tool clearance for bolted connections should be provided. In remote areas and where the use of tools is restricted, mounting brackets with self-locking nut plates may be used provided the nut plates are compatible with the environment. Sufficient access should be provided to all lubrication fittings to permit the use of standard USAF lubrication equipment without the need for adapters. Fuel and oil filters should be located for ready access. Bypass indicators and oil level indicators should be visible from access doors. Flexible plumbing lines should be used wherever practical to enhance removal and accessibility to other engine components. All accessories and components of the engine installation should be accessible for inspection, cleaning, and adjustment while installed in the aircraft.

The design should permit unobstructed adjustment over the entire range of component movement. Accessibility should not require removal of the engine, accessory gearbox, propeller, propeller gearbox, transmission, fluid tanks, fluid lines, or important parts of the aircraft structure.

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Service items accessible on a bare uninstalled engine should be accessible on the installed engine. The installation should be arranged to permit inspection, cleaning, adjustment, removal, and replacement of components and accessories with tools normally contained in a mechanic's tool kit; special tools should not be required. Examples of items which should be accessible are as follows:

a. Variable exhaust nozzle and control assembly

b. Variable inlet geometry components, controls, and sensors

c. Oil pressure relief valves

d. Oil tanks, pumps and oil level indicators

e. Oil filters/screens and bypass indicators

f. Oil coolers and temperature control valve

g. Fuel boost pumps

h. Afterburner fuel pump

i. Fuel filters and heaters

j. Fuel pressure relief valve

k. Fuel nozzles

l. Fuel control (engine and afterburner)

m. Fuel flow meter and divider

n. Automatic controls and actuators

o. Actuator motor brushes

p. Sump plugs and drain valves

q. Magnetic drain plugs

r. Ignitor plugs

s. Ignition excitors

t. Fire detection control units

u. Fire extinguisher spray rings

v. Fire extinguisher bottles

w. Engine mounts

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x. Gearbox mounts

y. Propulsion control linkages

z. Throttle boxes

aa. Ice detector and controls

bb. Inlet screen components and controls

cc. Inlet duct anti-icing components and controls

dd. Thrust reverser components and controls

ee. Bleed air ducting and valves

ff. Engine power takeoff (PTO) shaft and couplings

gg. Engine-mounted propeller governor

hh. Propeller unfeathering accumulator

ii. Engine borescope inspection provisions

jj. Engine thermocouples and harness

ll. Generators/alternators/CSDs

mm. Tachometer generator

nn. Hydraulic Pump

oo. Compressor inlet temperature sensor

pp. Fan speed sensor

REQUIREMENT LESSONS LEARNED

4.11 **Accessibility/maintainability.** The ready accessibility for the purposes of inspection, cleaning, and adjustment (as well as removal and replacement) of the various propulsion system components using standard tools shall be demonstrated with the engine installed.

VERIFICATION RATIONALE (4.11)

A demonstration of propulsion system accessibility and maintainability has been proven by experience to be the most appropriate verification means.

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VERIFICATION GUIDANCE

As a minimum the following accessibility/maintainability demonstrations should be accomplished:

a. All fuel, oil, and other fluid systems should be drained as required for the removal of the engine accessories.

b. Each engine accessory should be removed and reinstalled successively. This may require removal and replacement of access doors, accessory nose, cowling, etc.

c. Oil and hydraulic fluid systems should be closed or drained as required for the removal of the propeller, rotor, or fan assembly.

d. The propeller, rotor, or fan assembly should be removed and reinstalled.

e. Individual components of the propeller, rotor, or fan assembly (such as one main rotor blade) should be removed and reinstalled.

f. All items of inspection, cleaning, and adjustment that are normally performed during service use of the aircraft shall be accomplished.

g. Compressor cleaning operations, if applicable, should be accomplished.

h. Oil sampling techniques should be demonstrated.

i. Borescope inspection of internal parts at all inspection ports provided should be made.

j. A removal/replacement demonstration of thermocouple probes, harnesses, sensors, controls and accessories, airfoil stages (as applicable), and other enigne items designed for removal from the installed engine should be made.

VERIFICATION LESSONS LEARNED

3.12 **Reliability.** The reliability of the engine installation shall be compatible with the overall aircraft system. The mean time between failure (MTBF) for the engine installation designs and systems shall not be less than _____.

REQUIREMENT RATIONALE (3.12)

Engine installation reliability requirements are necessary to assure that the allocated reliabilities are compatible with the weapon system requirements.

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REQUIREMENT GUIDANCE

The contractor shall allocate the firm system numerical reliability requirements to appropriate equipment subdivisions. Requirements consistent with the allocations shall be imposed on the subcontractors and vendors. Contractor apportioned values shall be included in appropriate sections of procurement specifications, critial item specifications, and contract end item specifications. All allocated reliability values established by the contractor and included in contract end item specifications shall be subject to procuring activity approval (see MIL-STD-785).

REQUIREMENT LESSONS LEARNED

4.12 **Reliability.** The following analysis and test shall be accomplished to verify that allocated reliability requirements have been met.

TESTS	TEST HRS AND CYCLES	DEMONSTRATED RELIABILITY LEVELS

VERIFICATION RATIONALE (4.12)

Weapon system development experience has shown that the best way to obtain an acceptable level of confidence in propulsion system installation reliability is by test.

VERIFICATION GUIDANCE

Individual components should be exposed to a series of qualification tests to vendor specifications. Subsystem tests should be conducted in the laboratory and on the air vehicle. The composite of all testing constitutes the total reliability testing.

VERIFICATION LESSONS LEARNED

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3.13 Vulnerability. The vulnerability of critical propulsion components and accessories to threats commensurate with the aircraft mission shall be considered in the configuration of the propulsion system installation.

REQUIREMENT RATIONALE (3.13)

Aircraft survivability in a threat environment is highly dependent upon its ability to sustain damage. Design effort to reduce vulnerability enhances survivability.

REQUIREMENT GUIDANCE

To the maximum extent practicable, the accessories and components within the engine installation should be arranged or protected to ensure that the aircraft has the capability to return to its home base following a single hit from any weapon up to and including 23 millimeters. Fluid lines and essential components and wiring should be located to benefit from the inherent protection provided by nacelle structure and other less essential components and equipments. Redundant fluid lines, wiring, and components, when used, should be separated by a minimum of 18 inches. The separation distance need not be maintained if the redundant lines, etc, are separated by structure or other nonessential components which prevent the loss of both lines, etc, due to a single hit.

REQUIREMENT LESSONS LEARNED

4.13 Vulnerability. The survivability of the propulsion system installation shall be verified by _____.

VERIFICATION RATIONALE (4.13)

A combination of analysis, inspection and testing has proven to be the best approach to verifying compliance with the requirements of 3.13.

VERIFICATION GUIDANCE

Inspection and analysis are the primary means of verifying that adequate consideration has been given to the vulnerability/survivability of the propulsion system design.

VERIFICATION LESSONS LEARNED

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3.14 Infrared (IR) signature and radar cross section (RCS). The installed propulsion system shall meet the allocated IR and RCS requirements of the air vehicle specification.

REQUIREMENT RATIONALE (3.14)

A weapon system's capability to perform an intended mission may be highly dependent upon the ability to minimize tracking and surveillance signatures such as IR and RCS.

REQUIREMENT GUIDANCE

The weapon system's mission and threat environment will dictate the air vehicle and subsystem requirements for reduced observables. Propulsion system requirements must be derived from overall air vehicle requirements.

REQUIREMENT LESSONS LEARNED

4.14 Infrared (IR) signature and radar cross section (RCS). The propulsion system IR signature and radar cross section shall be determined by test.

VERIFICATION RATIONALE (4.14)

Experience has demonstrated that values which define IR signature and RCS of the propulsion system are quantified with the highest level of confidence when obtained by test.

VERIFICATION GUIDANCE

Tests should be run to determine the peak radiation in the IR portion of the electromagnetic spectrum for the complete installed propulsion system. Radar reflectivity measurements should also be made. Radiation and reflectivity should be measured over an angular range in both azimuth and evaluation and as appropriate in the forward and aft hemispheres.

VERIFICATION LESSONS LEARNED

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3.15 Noise. Near and far field noise levels for the installed propulsion system shall not exceed the levels specified by _____. Maximum noise exposure limits for Air Force ground personnel in terms of sound pressure levels and exposure times shall not exceed _____.

REQUIREMENT RATIONALE (3.15)

Propulsion system noise represents an annoyance to civilian communities and a health hazard to military personnel and therefore should be controlled.

REQUIREMENT GUIDANCE

Noise reduction, while desirable, must remain secondary to performance requirements. No noise limits are defined since such limits could place constraints on necessary performance. Propulsion system noise limits should be defined on a program by program basis. Propulsion system noise levels are estimated when the Air Force procures new systems. Noise levels should include consideration of crew environment aircraft structural requirements, maintenance and ground operation. Combat aircraft are usually not required to meet any standards. Transport type aircraft should be required to meet flight noise requirements set forth in the Federal Aviation Regulations (FAR) Part 36. This requirement for transport aircraft is outlined in Air Force Regulation (AFR) 80-36. Transport aircraft are exempt from FAR Part 36 only if it is determined that compliance would inhibit mission success.

AFR 161-35 dictates the maximum noise exposure limits for Air Force ground personnel in terms of sound pressure levels and exposure times. It also specifies precautions for those who must work above those limits.

REQUIREMENT LESSONS LEARNED

Noise from aircraft engines being tested or operated in the vicinity of populated areas has been responsible for complaints from the public. These complaints have imposed an additional work load on the Air Force and have, in some cases, resulted in restricted operation of Air Force systems.

4.15 Noise. Near and far field noise tests shall be conducted on the installed propulsion system. Compliance with the specified noise exposure limits for ground personnel shall be verified by measurement.

VERIFICATION RATIONALE (4.15)

Noise testing has proven by experience to be the only effective way of determining installed propulsion system noise levels.

VERIFICATION GUIDANCE

An installed propulsion system should be subjected to a comprehensive ground noise survey as specified in AFR-80-36 and AFR 161-35.

VERIFICATION LESSONS LEARNED

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3.16 Environmental extremes. Environmental extremes in which the propulsion installation components and systems must operate satisfactorily, include the following:

Natural Environment	Requirement
Altitude	
Temperature	
Salt fog	
Humidity	
Rain	
Dust	
Fungus resistance	

Induced Environment	Requirement
Gunfire vibration	
Shock	
Explosive atmosphere	
Acceleration	
Electromagnetic	

REQUIREMENT RATIONALE (3.16)

The environmental conditions under which the engine installation must function are specified to assure that the capability exists to operate effectively in all environments likely to be encountered worldwide.

REQUIREMENT GUIDANCE

Refer to MIL-STD-210 for guidance.

REQUIREMENT LESSONS LEARNED

4.16 Environmental extremes. All components within the propulsion installation system shall be subjected to the following environmental tests:

VERIFICATION RATIONALE (4.16)

Components of the propulsion installation should be subjected to the specified tests in order to demonstrate their compatibility with the environmental requirements applicable to the propulsion system installation.

VERIFICATION GUIDANCE

Methods of environmental testing are presented in MIL-STD-810. Test requirements applicable to the weapon system should apply to the propulsion system installation.

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VERIFICATION LESSONS LEARNED

3.17 Interface control. The airframe and engine physical, functional, and performance interfaces shall be defined and controlled through an Interface Control Document.

REQUIREMENT RATIONALE (3.17)

Airframe and engine interfaces must be defined and controlled since interface changes by either contractor will impact the other.

REQUIREMENT GUIDANCE

The Interface Control Document (ICD) is usually prepared by the airframe contractor with inputs from the engine manufacturer. This document defines all physical, functional and performance interfaces and an administrative procedure whereby these interfaces can be controlled and ongoing coordination between the airframe and engine contractors can be facilitated. The ICD typically includes all interfaces described in this specification as well as interfaces of the bleed air system and fuel system and those interfaces that might exist with airframe-mounted lubrication system components (i.e., oil tanks, fuel/oil, air/oil heat exchangers, etc.).

REQUIREMENT LESSONS LEARNED

4.17 Interface control. Definition and control of the airframe and engine physical, functional and performance interfaces shall be verified by inspection.

VERIFICATION RATIONALE (4.17)

An inspection of the airframe and engine interfaces and the procedures to control these interfaces has proven sufficient to establish that the contractor is paying proper attention to this critical area.

VERIFICATION GUIDANCE

Contractor control of airframe and engine interfaces should be established as early as possible. This verification is best accomplished by incremental inspections of the evolving interfaces throughout the design process.

VERIFICATION LESSONS LEARNED

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3.18 Safety. Safety requirements for the engine installation shall include the following: _____.

REQUIREMENT RATIONALE (3.18)

Safety requirements for the engine installation are necessary to assure that hazards to the air vehicle, crew and ground personnel have been reduced to the maximum extent possible.

REQUIREMENT GUIDANCE

Propulsion system warning indicators and fire extinguishing controls should be provided in the cockpit per AFSC DH 2-2, DN 2A8. Electrical equipment and wiring components located in the propulsion system installation should be explosion proof. Design removable cowlings and doors to be of reasonable dimensions and weight for ease of handling by ground personnel and ensure that no sharp corners exist which may be hazardous. Design cowlings, doors, and panels so they do not separate from the nacelle in the event of a high pressure bleed air line rupture or experience structural damage which would result in a situation impairing safe operation of the aircraft. Provide pylon and nacelle cowling with pressure blowout doors where high pressure bleed lines are located.

Refer to AFSC DH 1-6, Section 3D for additional guidance.

REQUIREMENT LESSONS LEARNED

4.18 Safety. The safety requirements for the engine installation shall be verified by: _____.

REQUIREMENT	VERIFICATION METHOD

VERIFICATION RATIONALE (4.18)

Various methods may be required to verify the specified safety requirements. The most appropriate method based on historical experience and program requirements should be defined.

VERIFICATION GUIDANCE

Analysis and testing are methods typically used to verify the propulsion system.

VERIFICATION LESSONS LEARNED

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-87239(USAF)	Installation, Engine

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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